FAKULTA APLIKOVANÝCH VĚD
ZÁPADOČESKÉ UNIVERZITY
V PLZNI

Edgebreaker

Komprese trojúhelníkových sítí

Jakub Vokoun

Fakulta aplikovaných věd ZČU

javok@students.zcu.cz

2026-11-04

1. Motivace

1.1 Dobový kontext

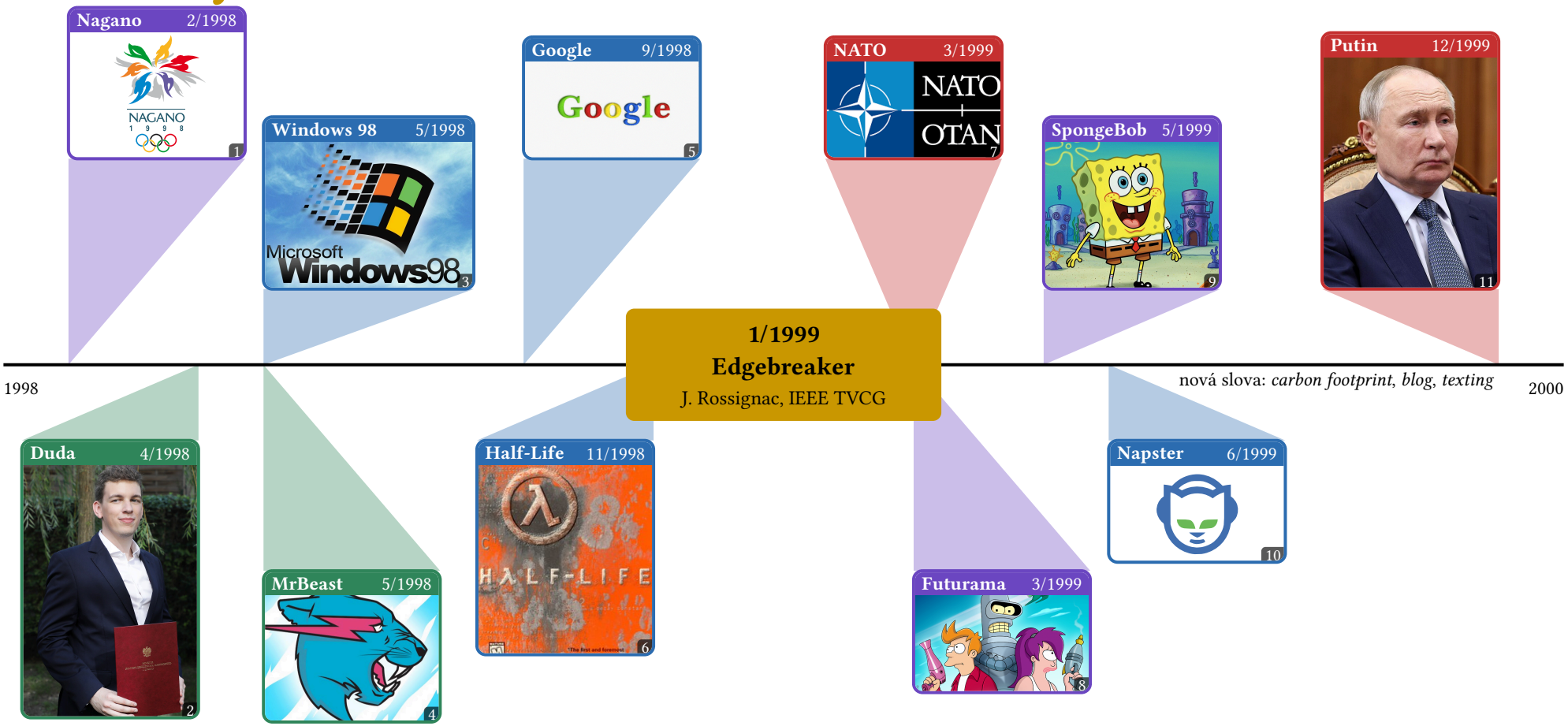


Figure 1: Dobový kontext vzniku algoritmu Edgebreaker.

1.1 Dobový kontext



Figure 2: Jarek Rossignac, zdroj: http://kelvinkuo.com/wp-content/uploads/2010/03/KKK_6258.jpg

2. Definice problému

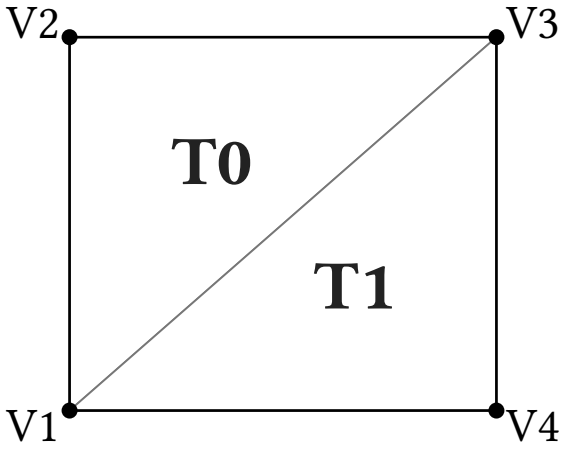


Figure 3: Trojúhelníková mesh.

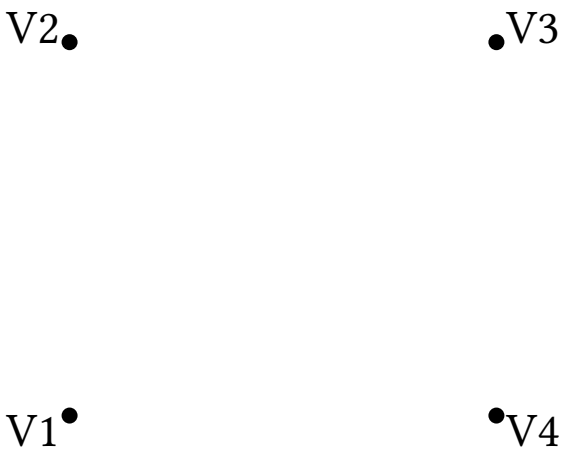


Figure 3: Trojúhelníková mesh.

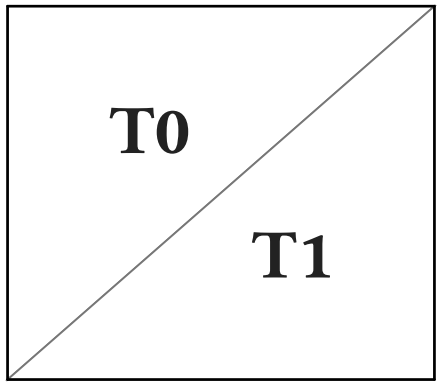


Figure 3: Trojúhelníková mesh.

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

CLERS řetězec

2.2 Jak schovat informaci?

pořadí vertexů	CLERS řetězec
[V1 V2 V3]	CLRSLRCLEE
[V1 V3 V2]	
[V2 V1 V3]	
[V3 V1 V2]	
[V2 V3 V1]	
[V3 V2 V1]	

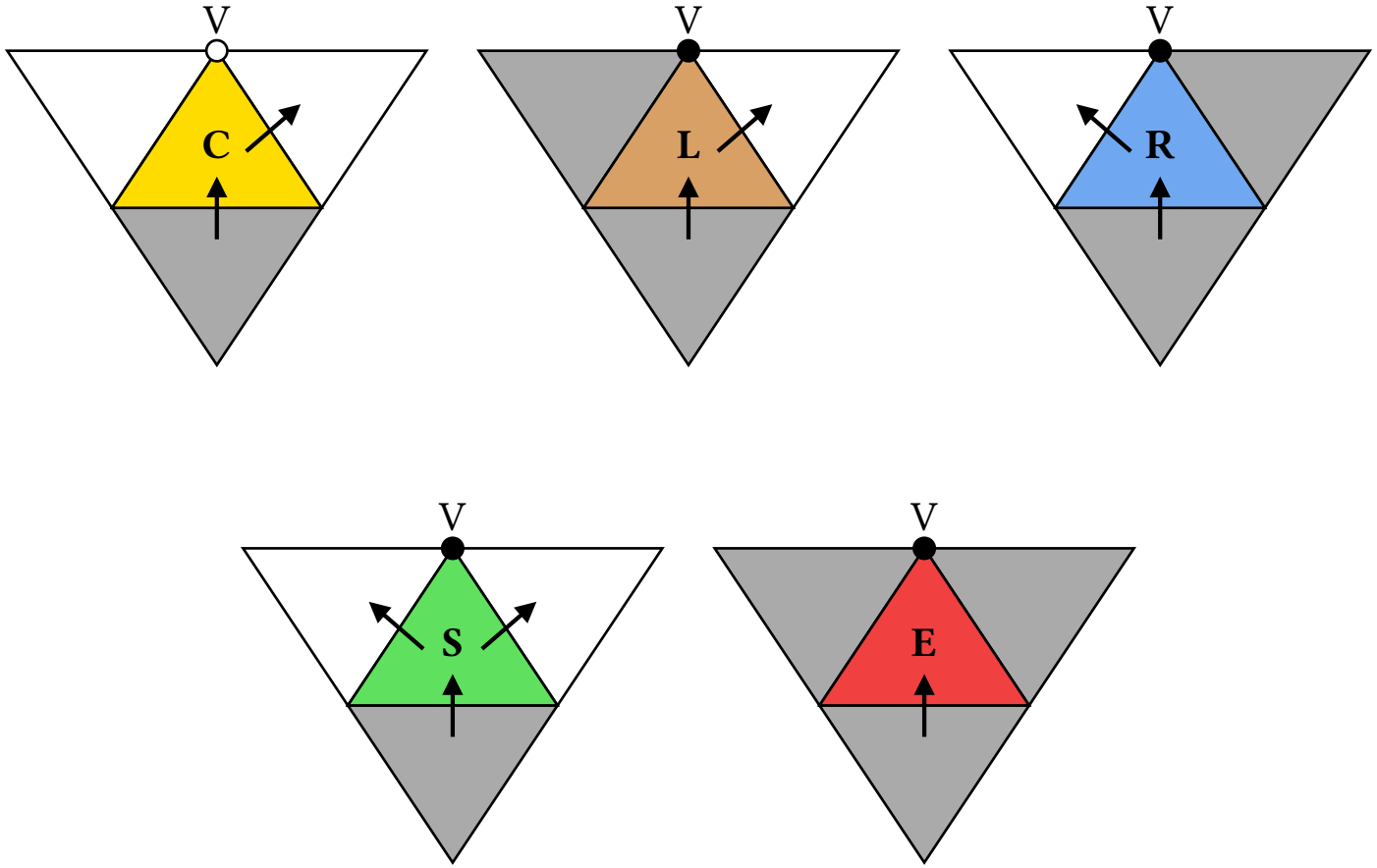


Figure 4: Základní CLERS operace.

3. Komprese

3.1 Pravidla

History =

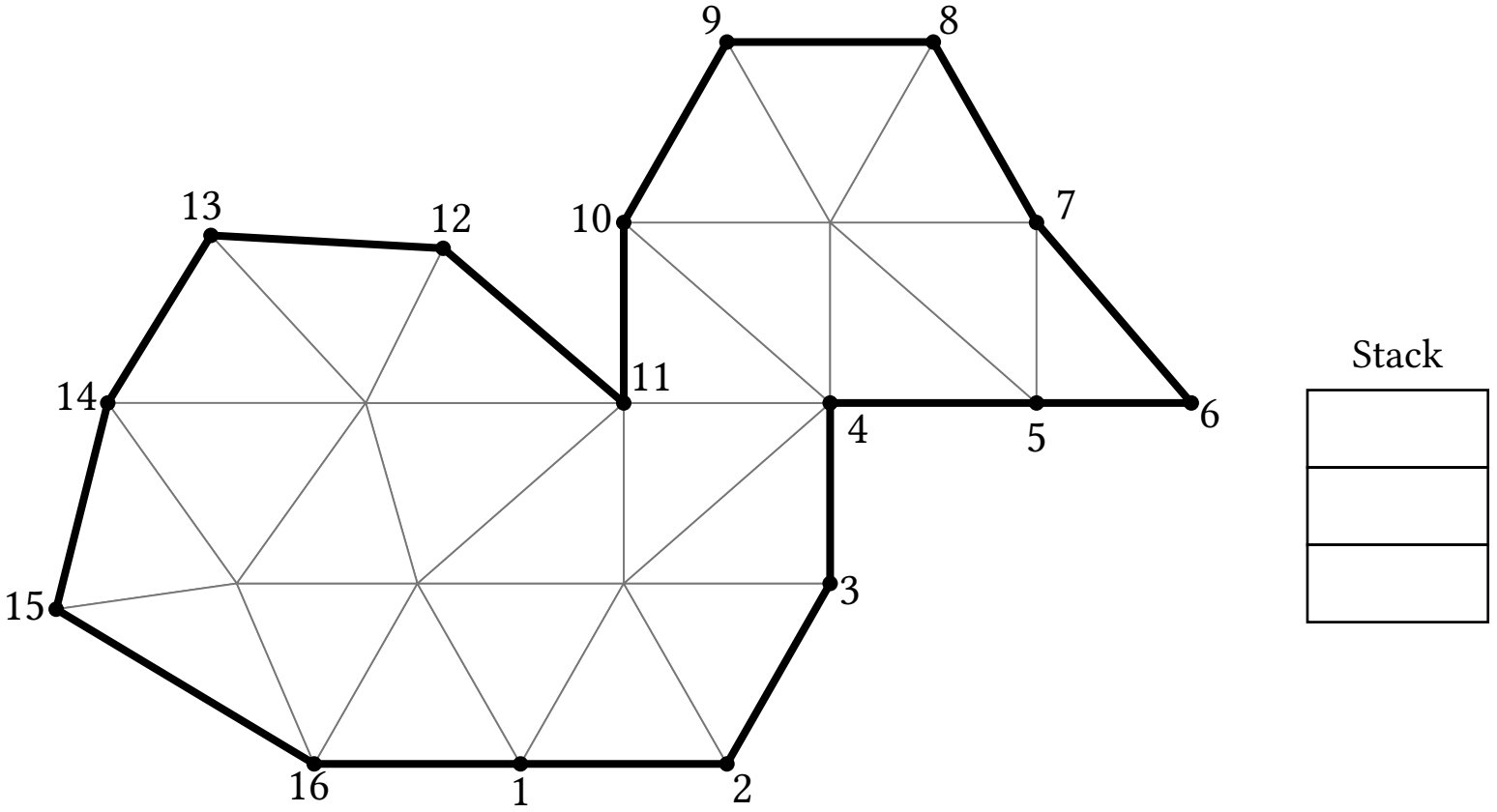


Figure 5: Jednoduchá komprese.

History = C

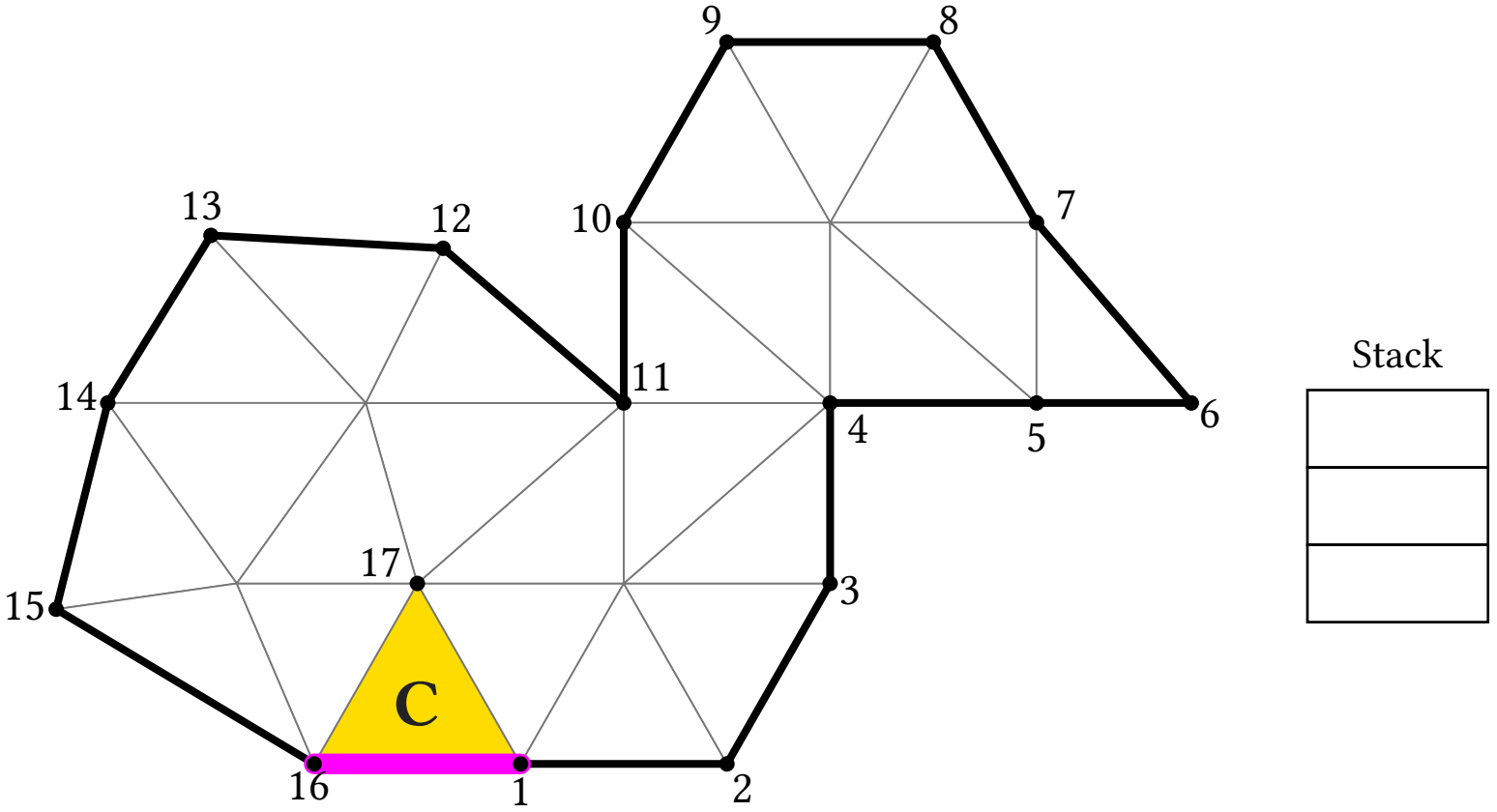


Figure 5: Jednoduchá komprese.

History = CC

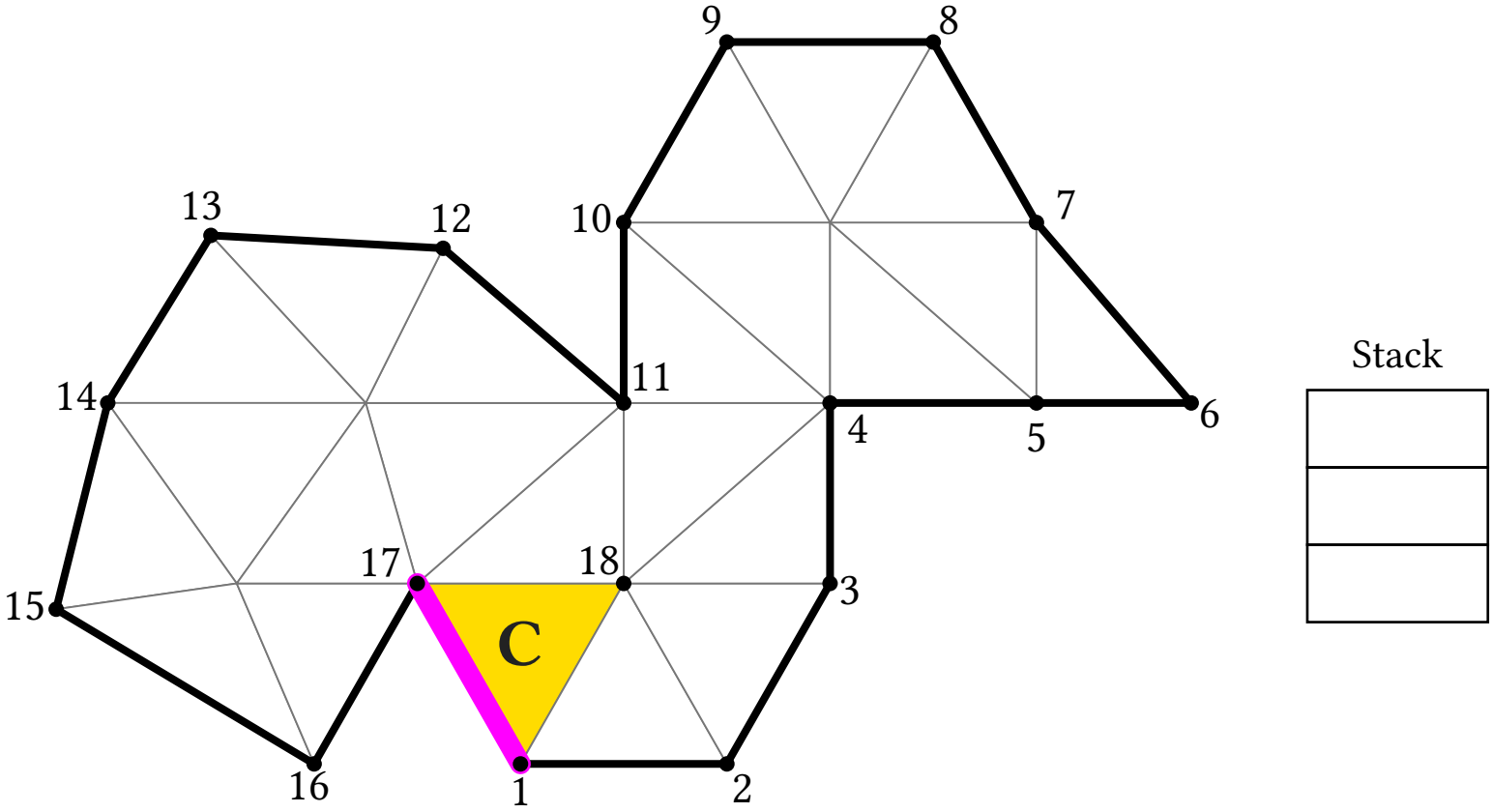


Figure 5: Jednoduchá komprese.

History = CCR

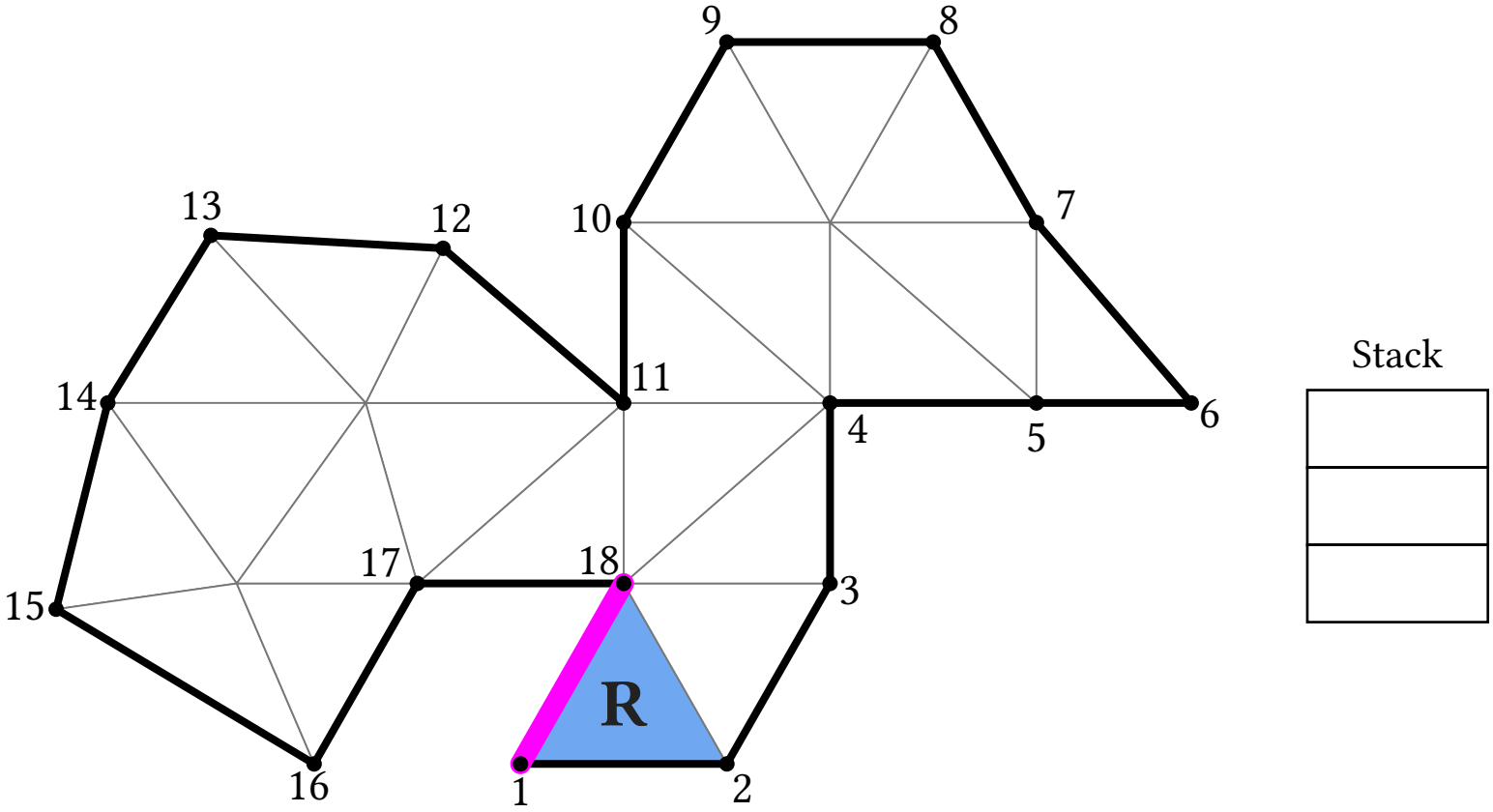


Figure 5: Jednoduchá komprese.

History = CCRR

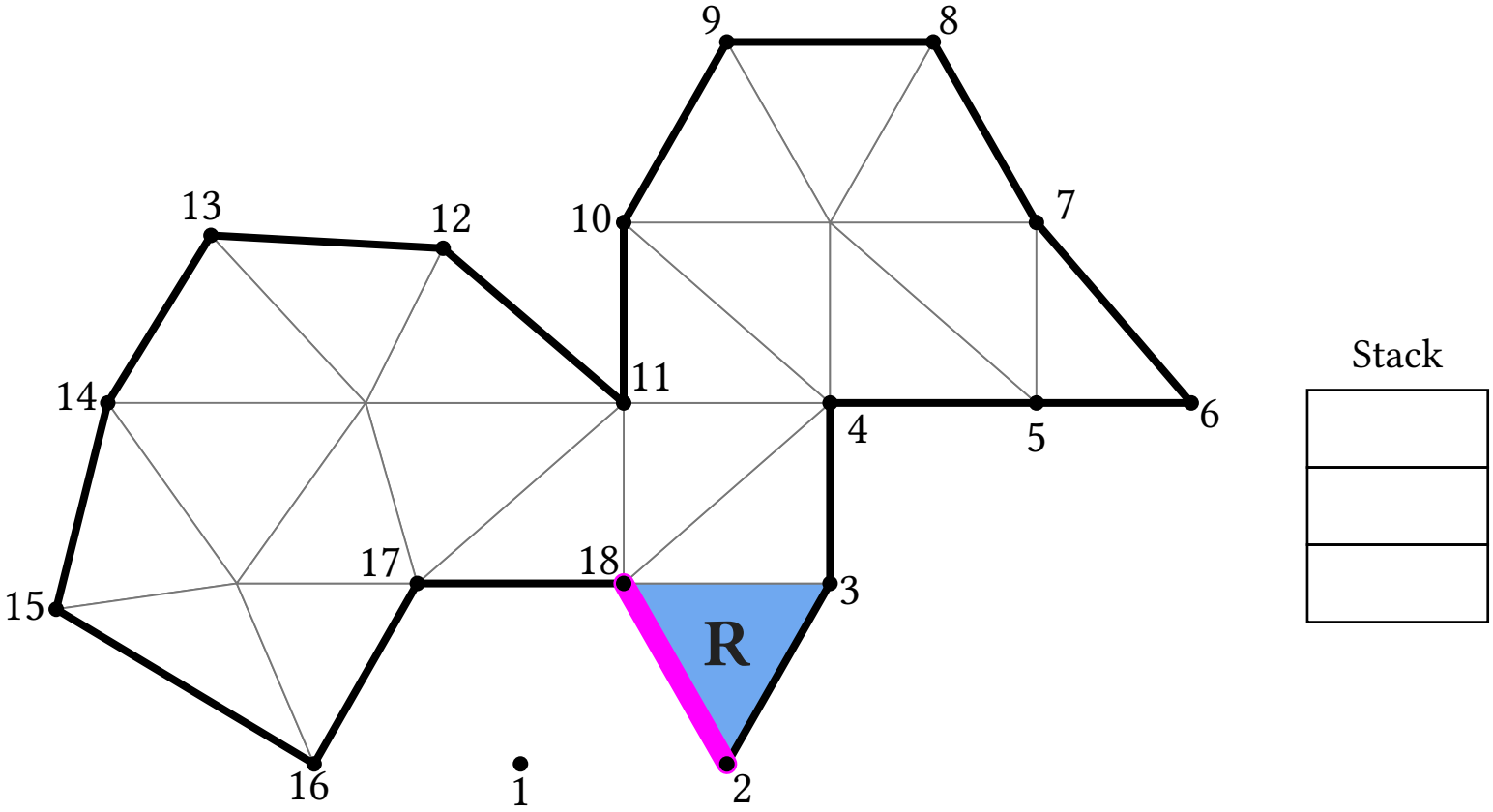


Figure 5: Jednoduchá komprese.

History = CCRRR

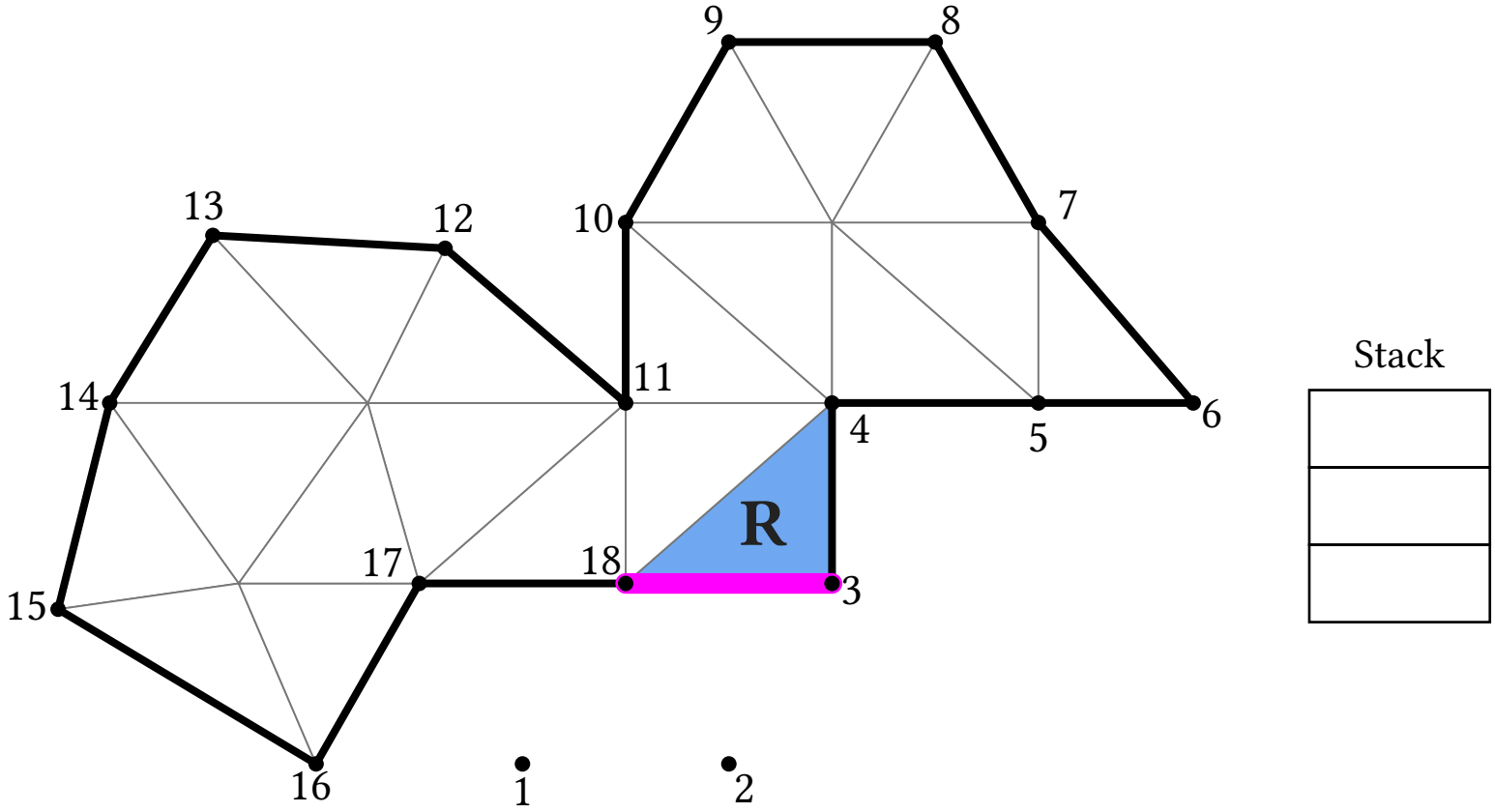


Figure 5: Jednoduchá komprese.

History = CCRRRS

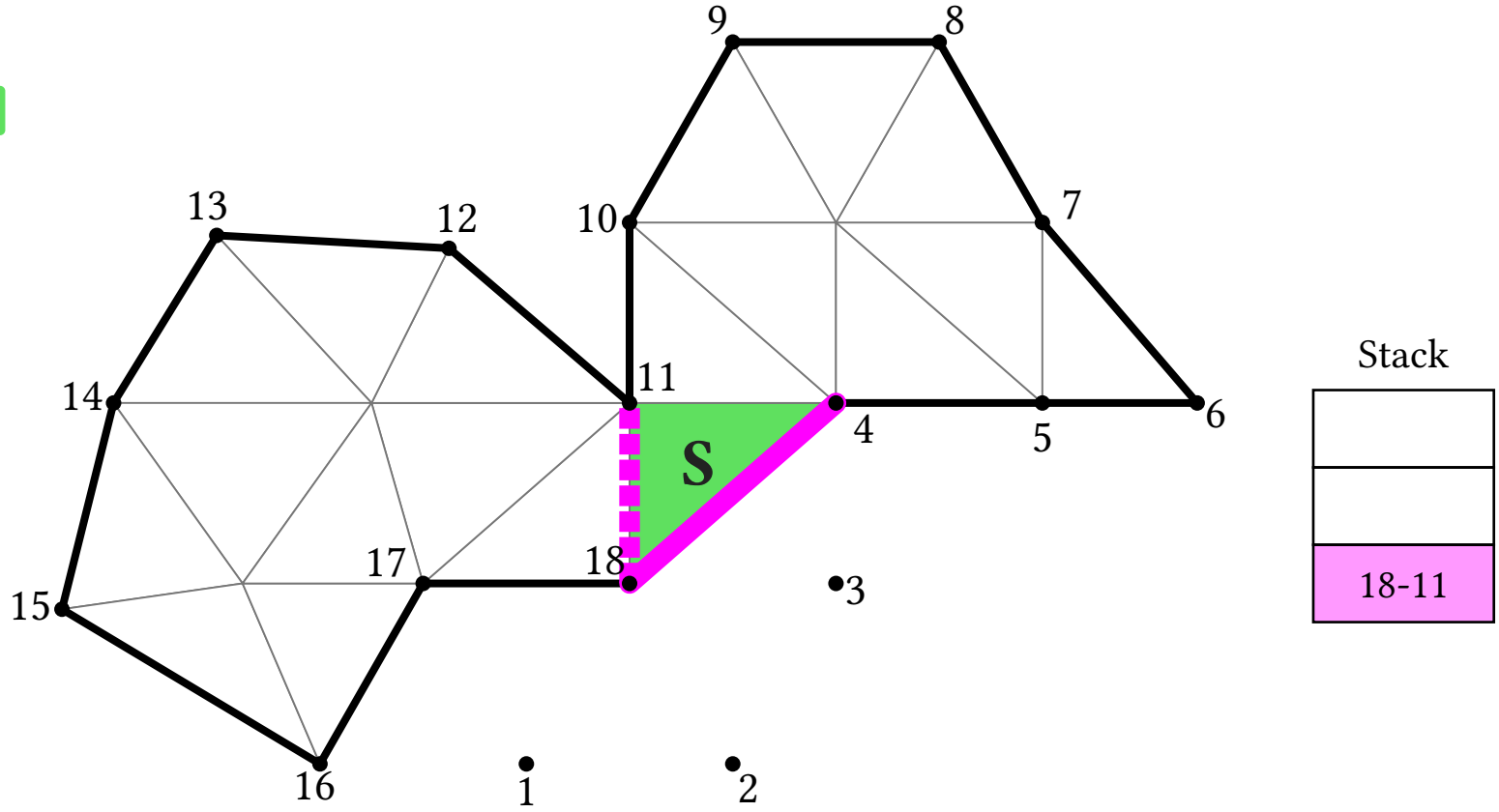


Figure 5: Jednoduchá komprese.

History = CCRRRSL

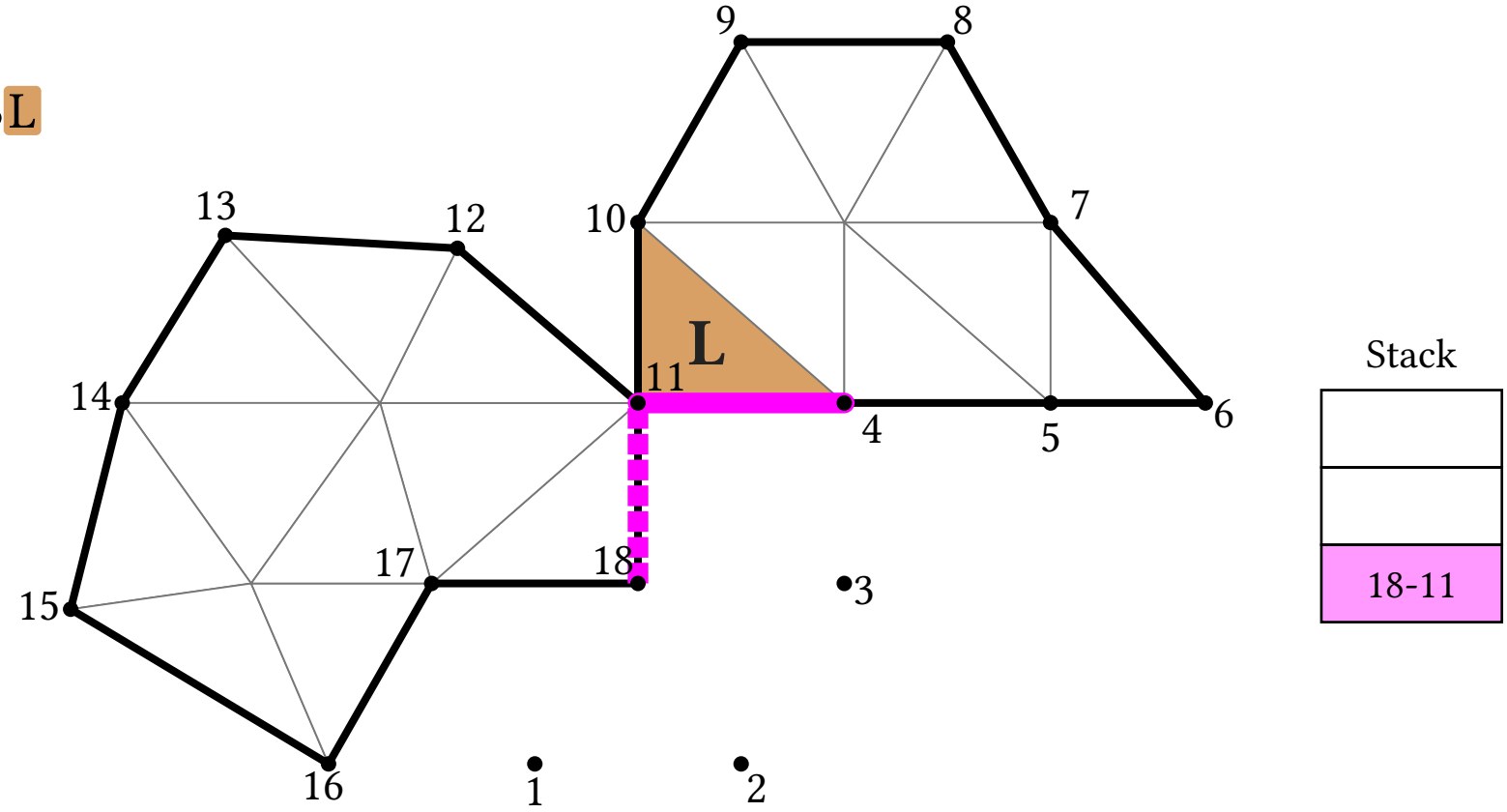


Figure 5: Jednoduchá komprese.

History = CCRRRSLC

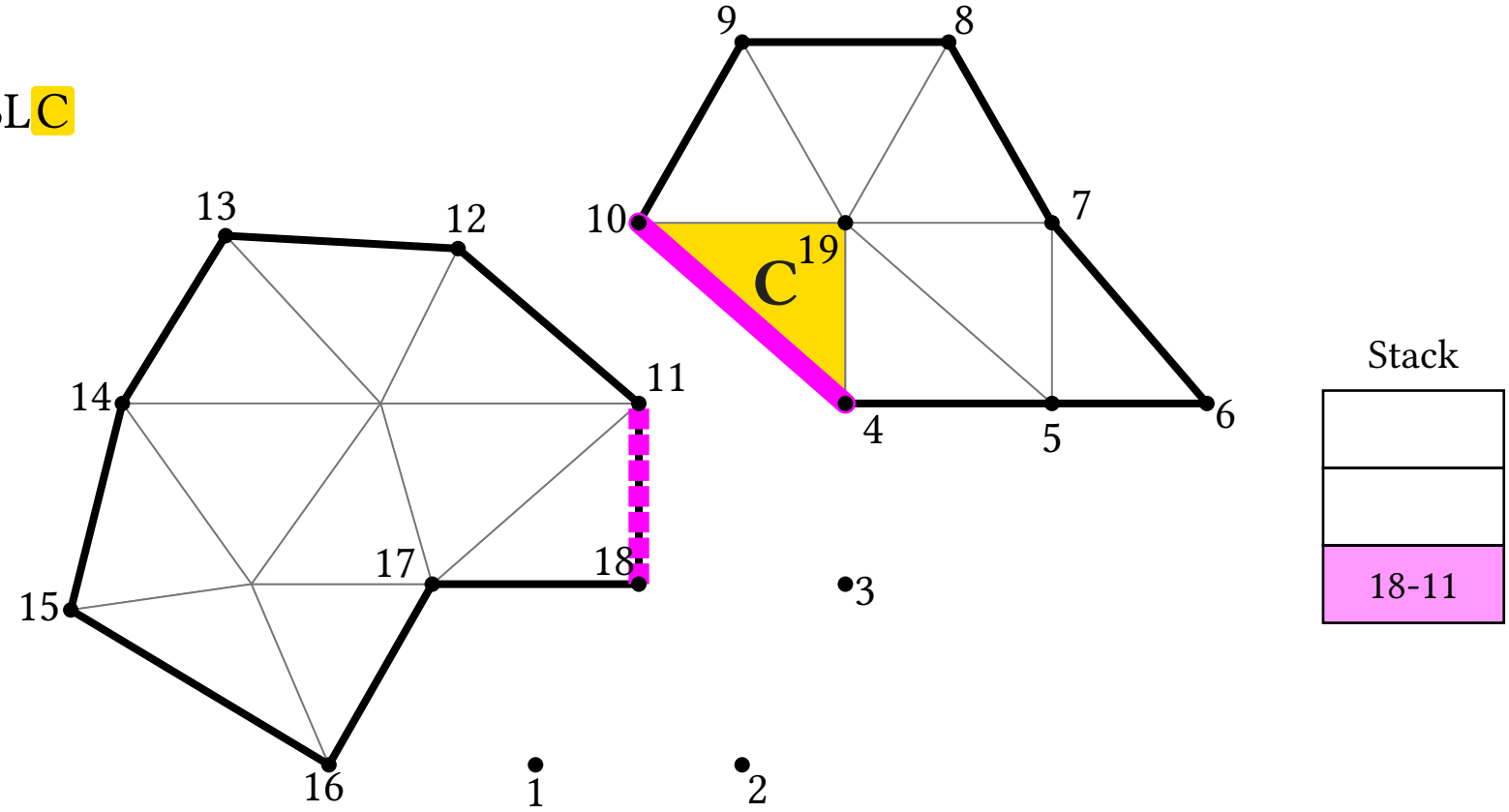


Figure 5: Jednoduchá komprese.

History = CCRRRSLCR

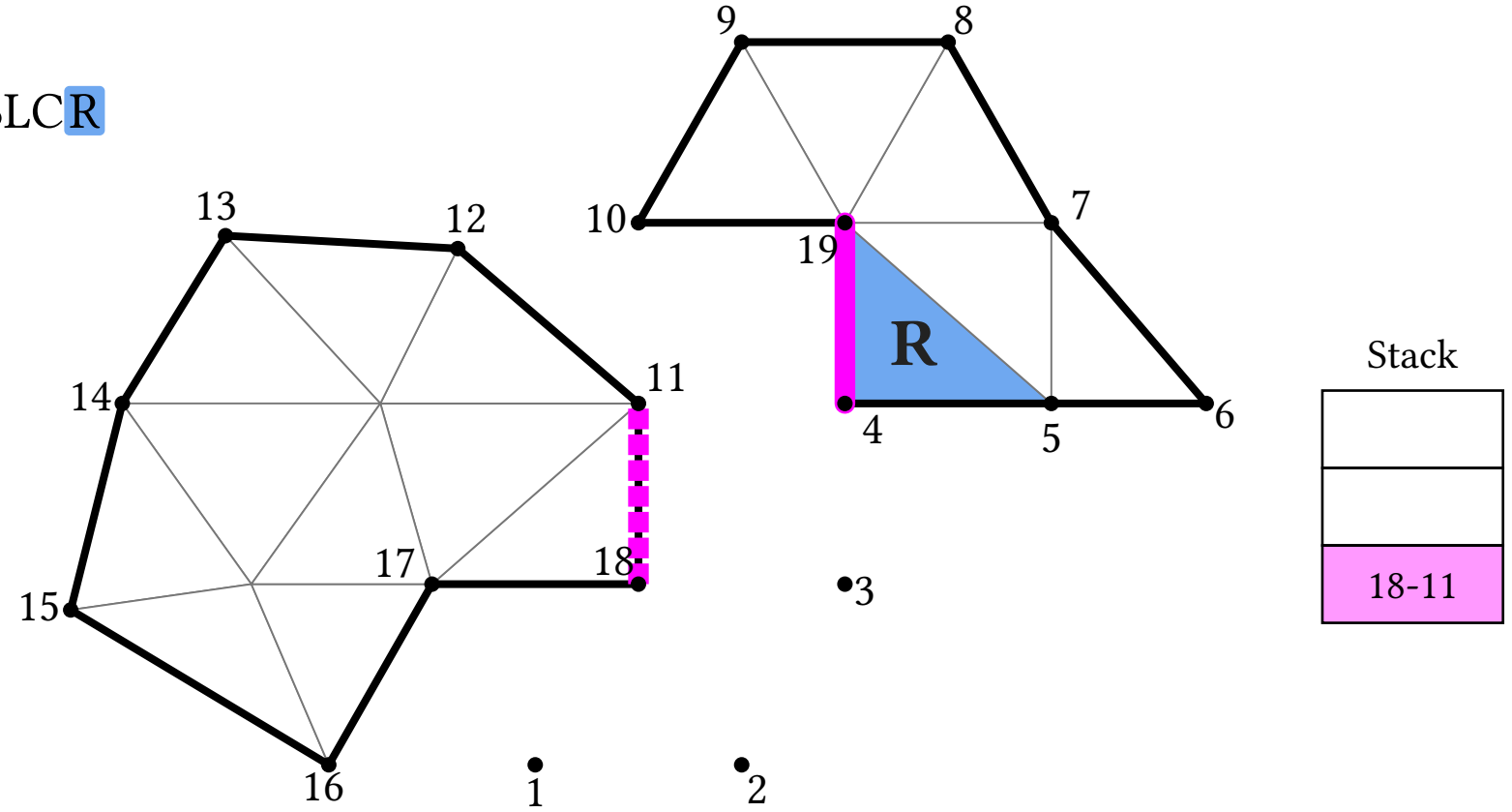


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRS

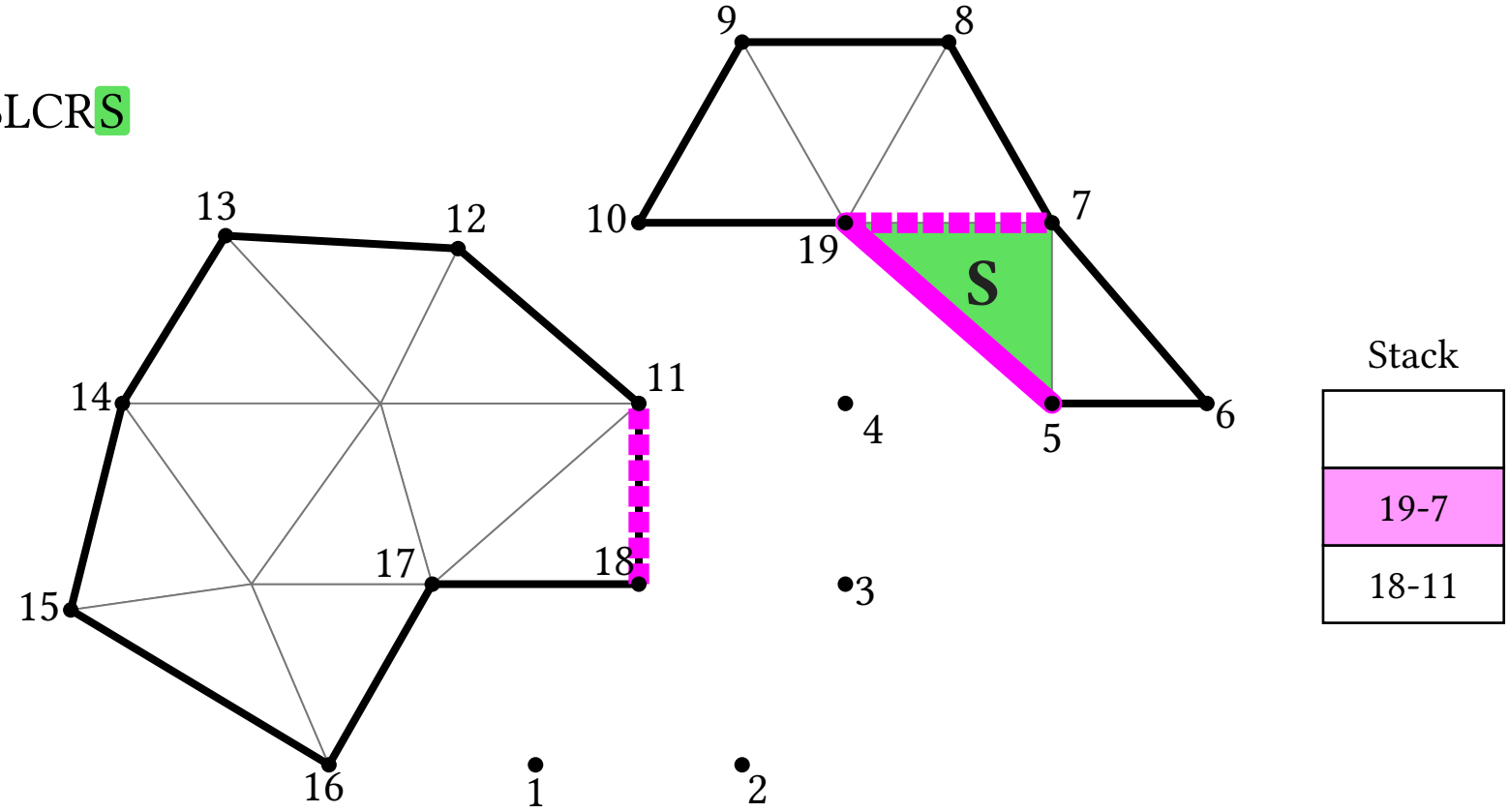


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSE

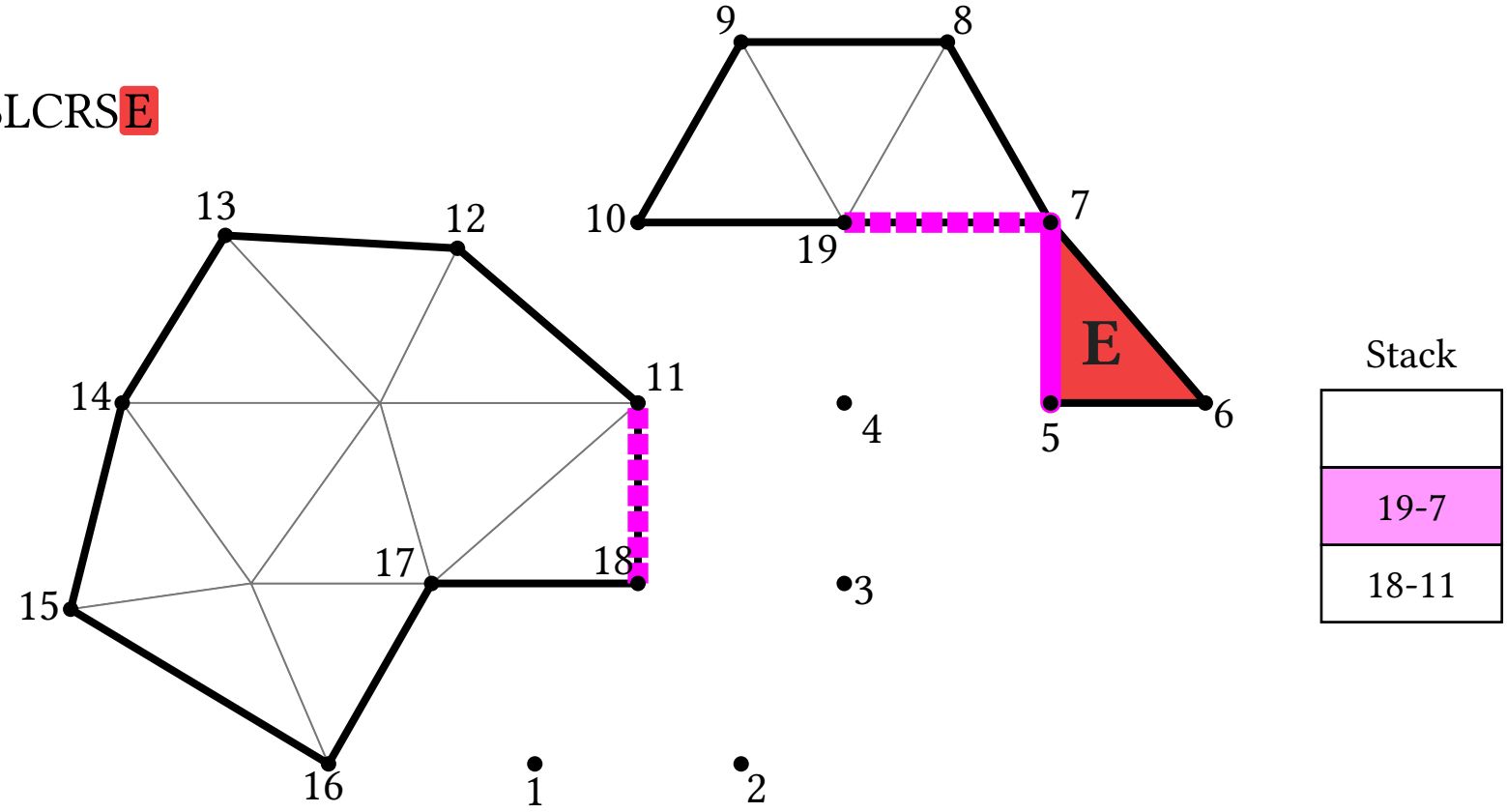


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSER

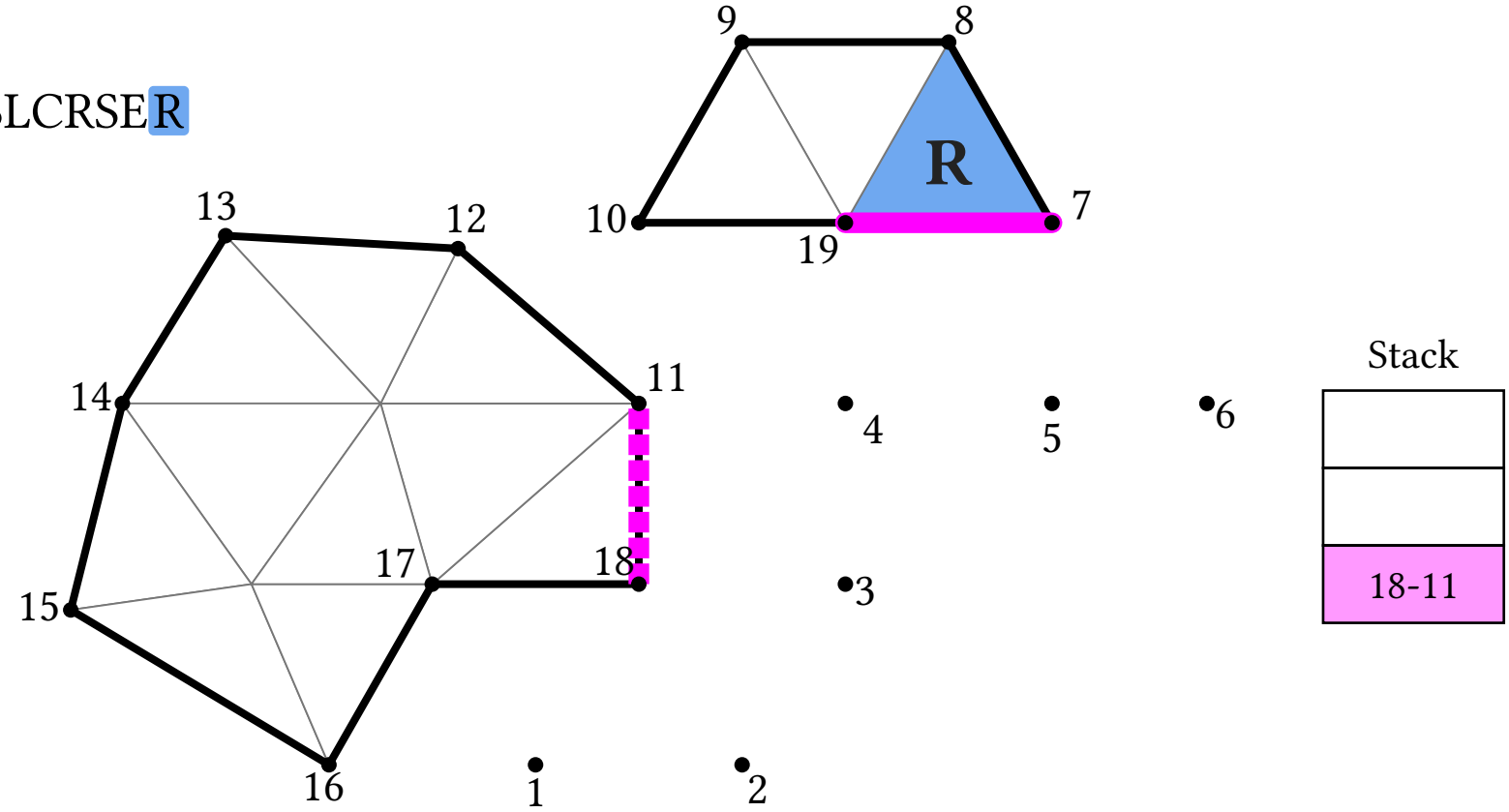


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERR

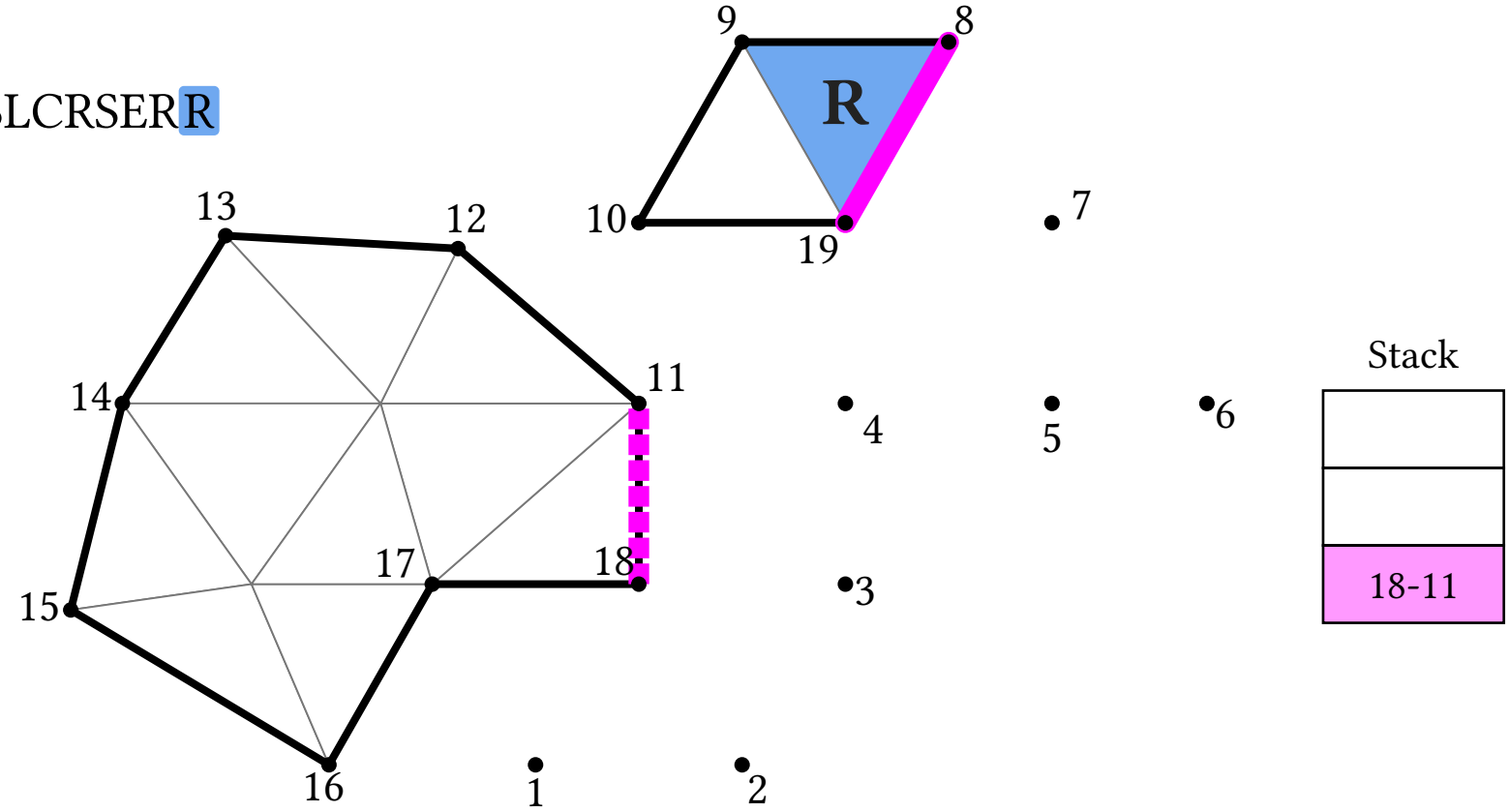


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRE

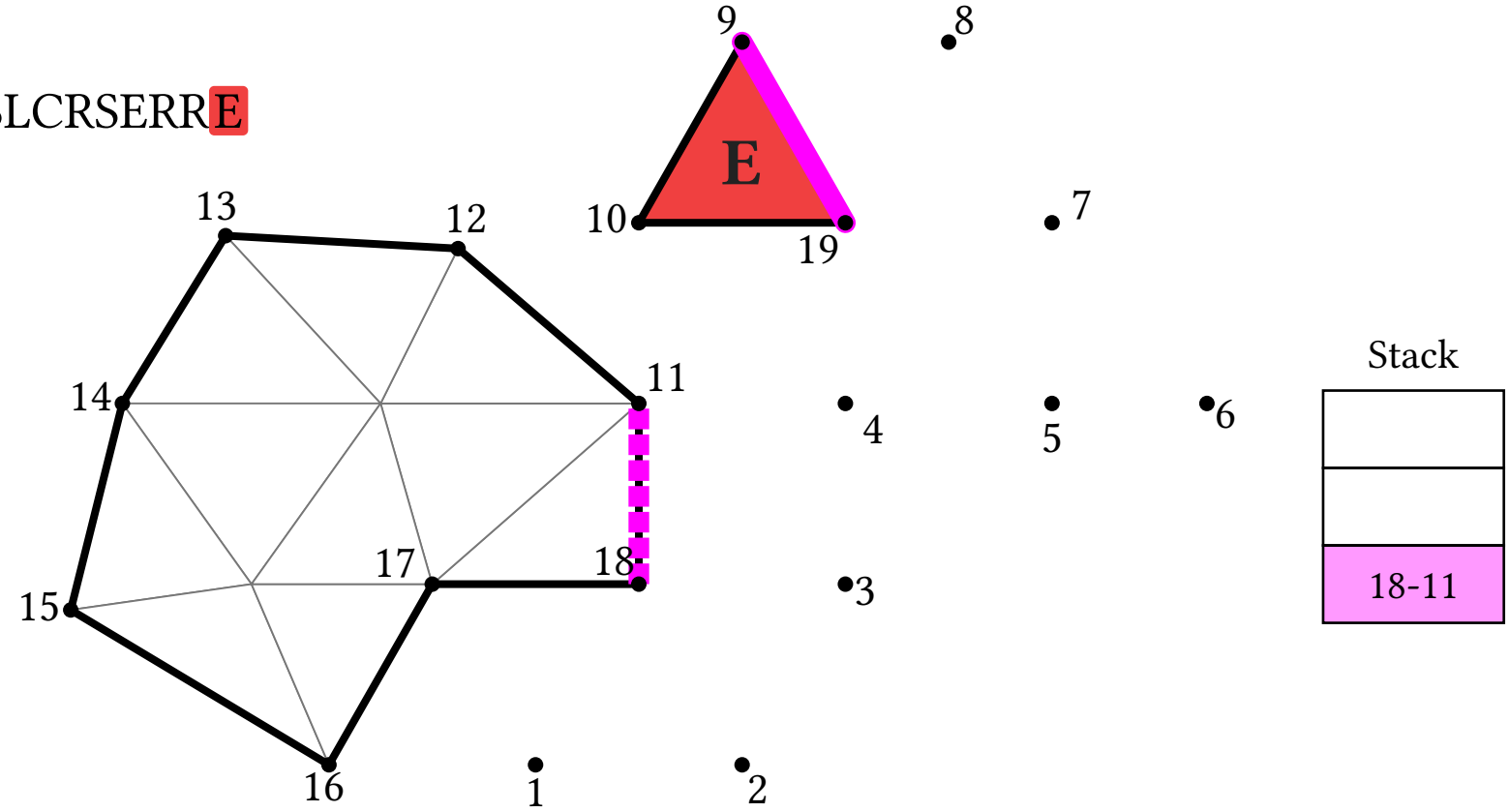


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERREL

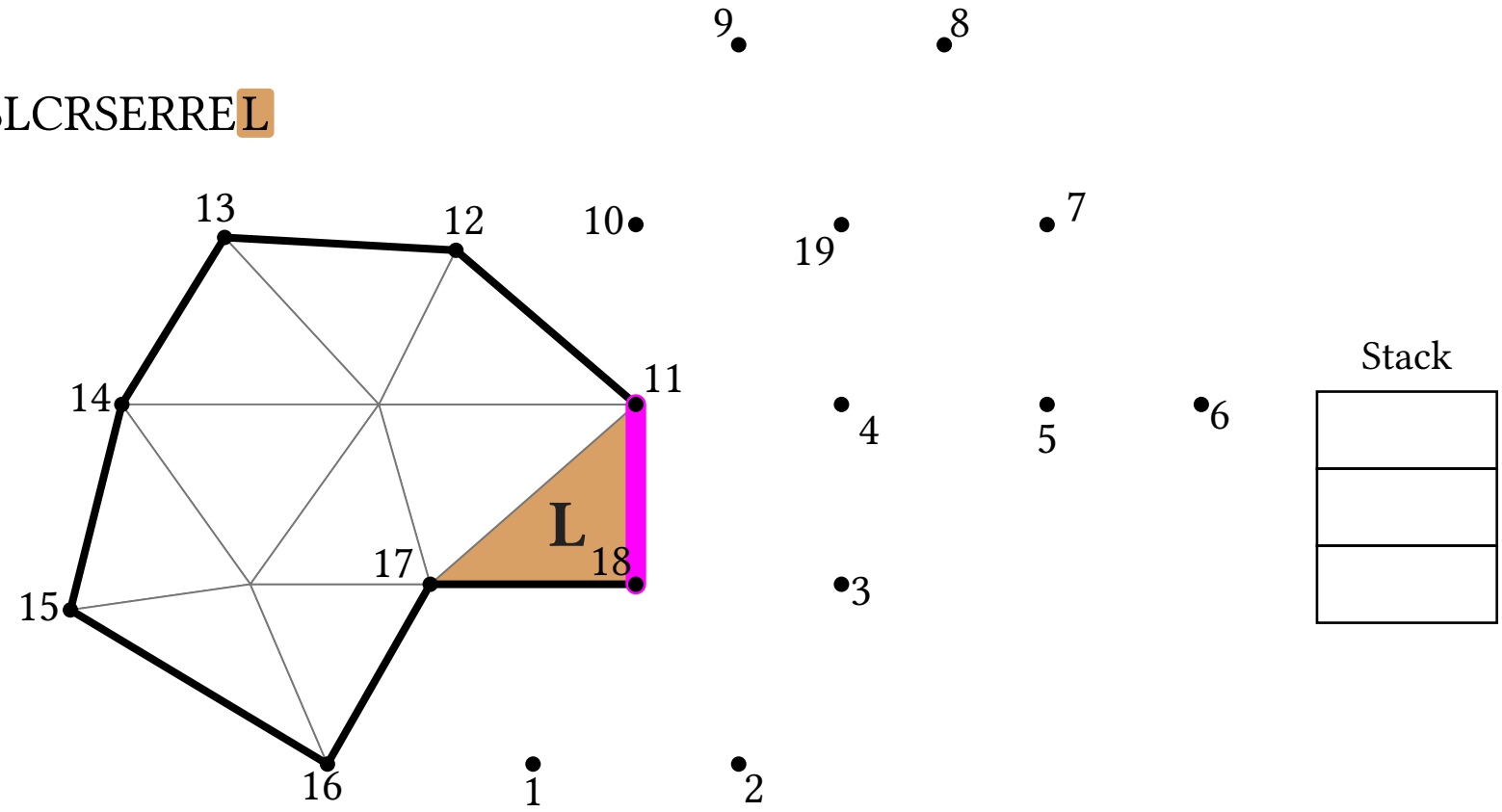


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERREL**C**

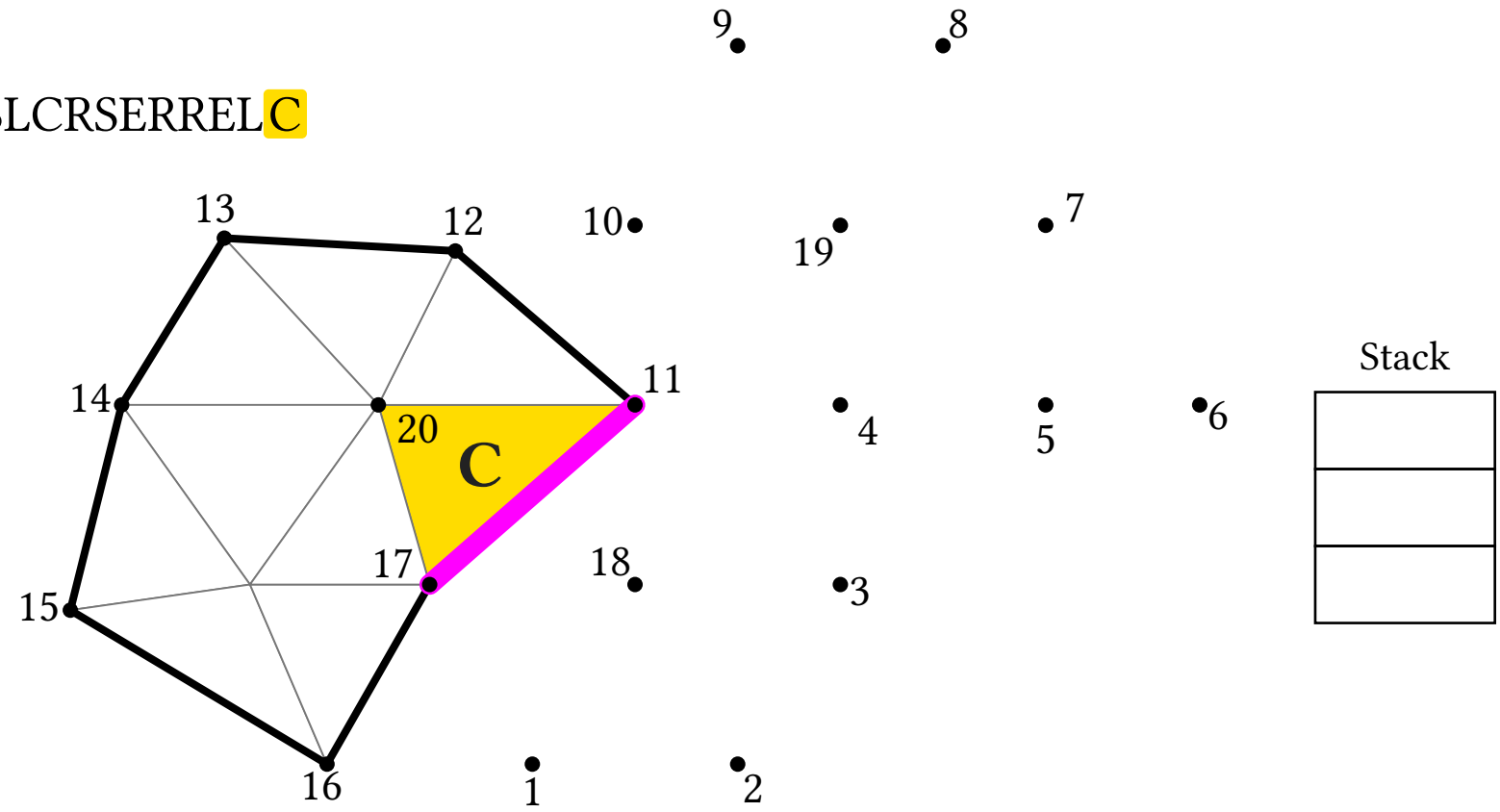


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCR

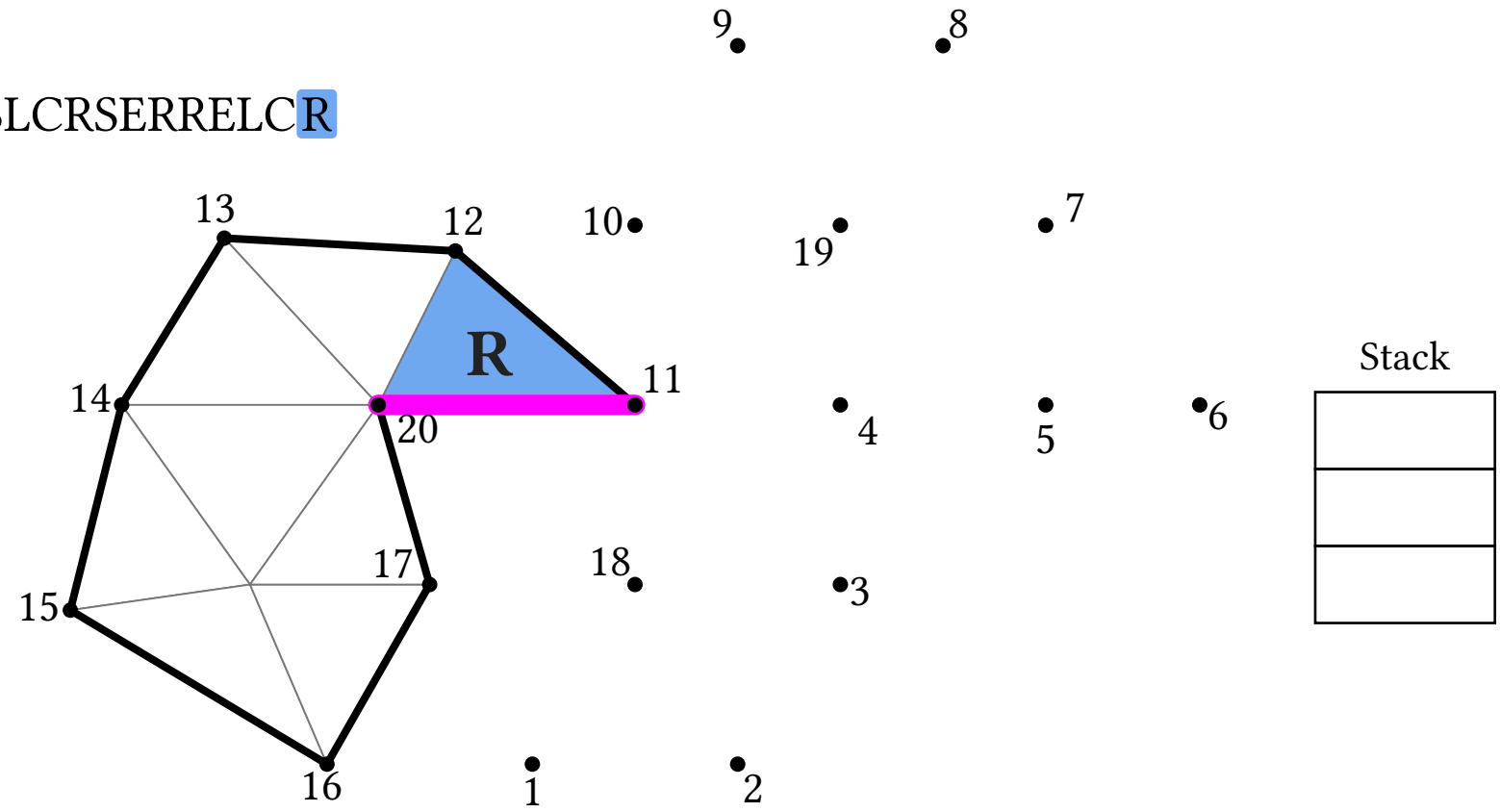


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRR

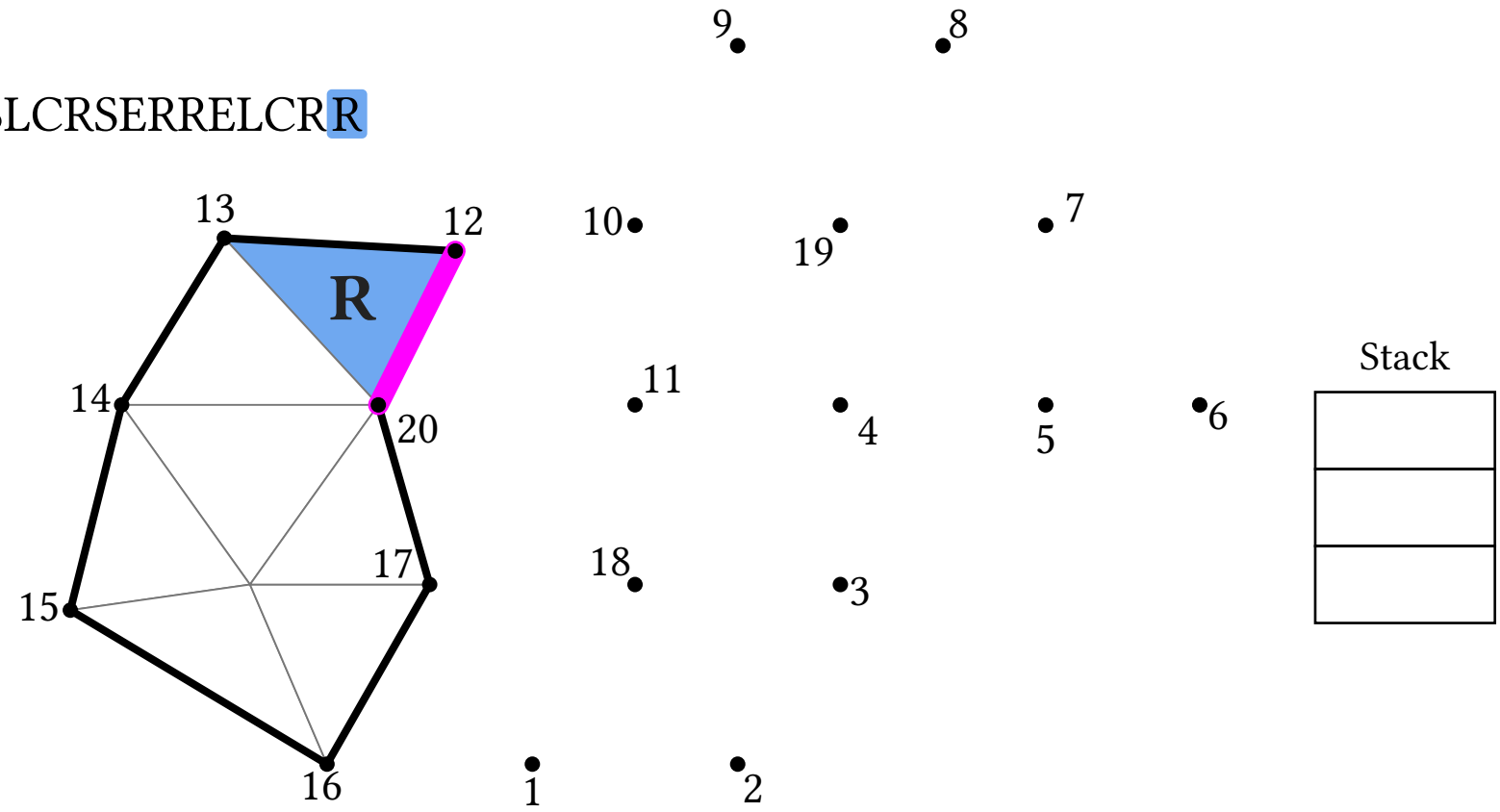


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRR

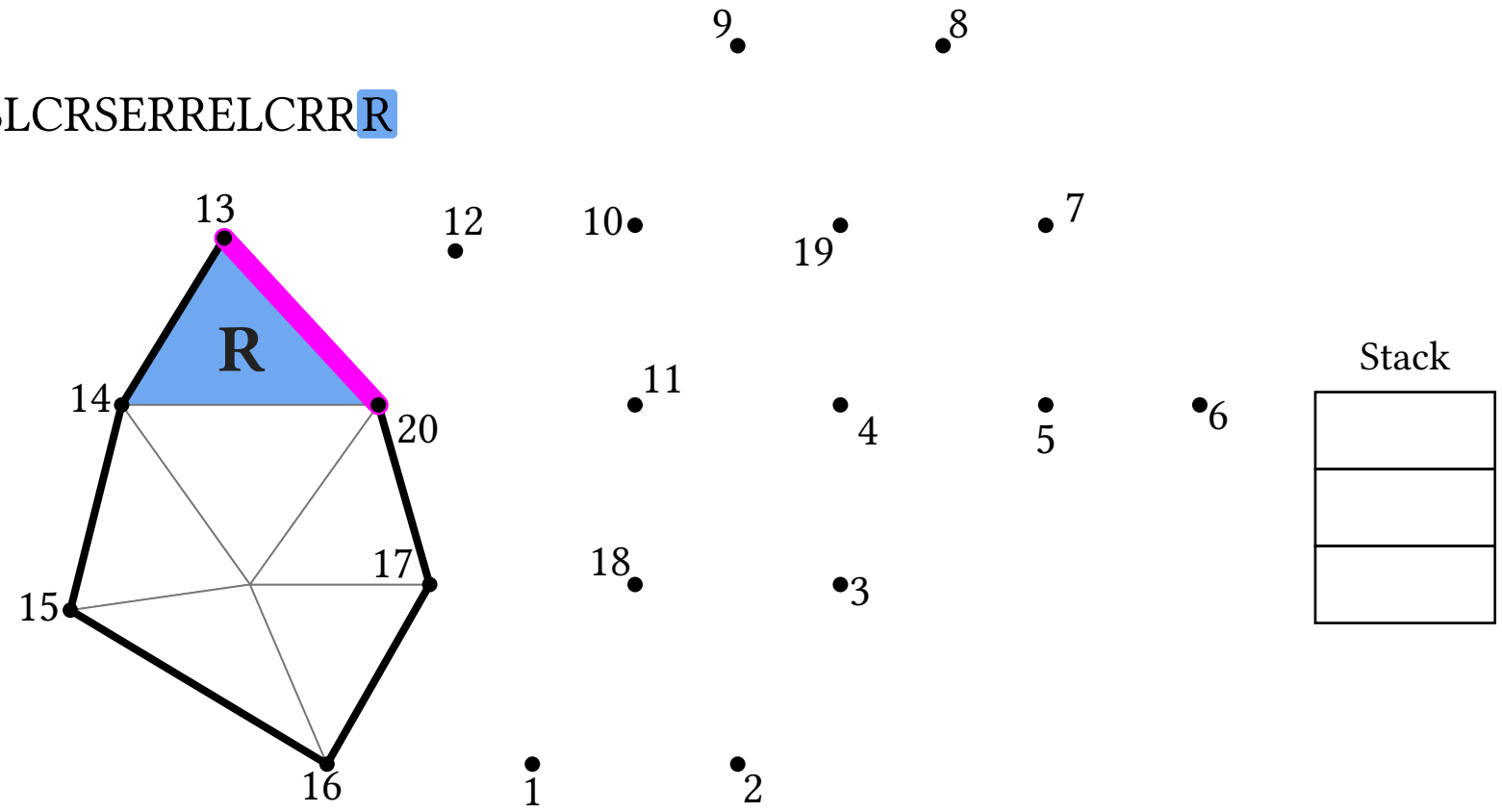


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRC

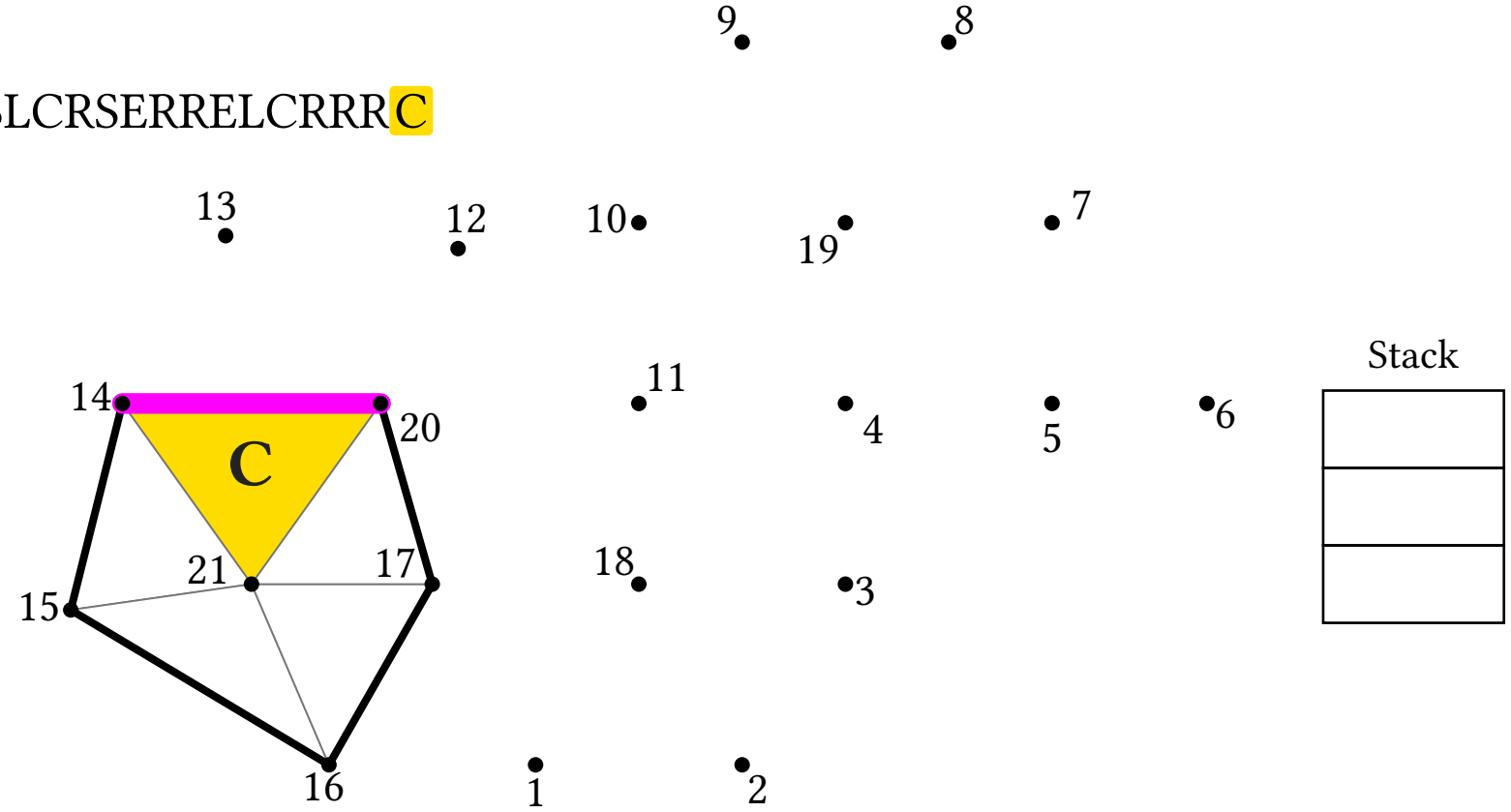


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCR

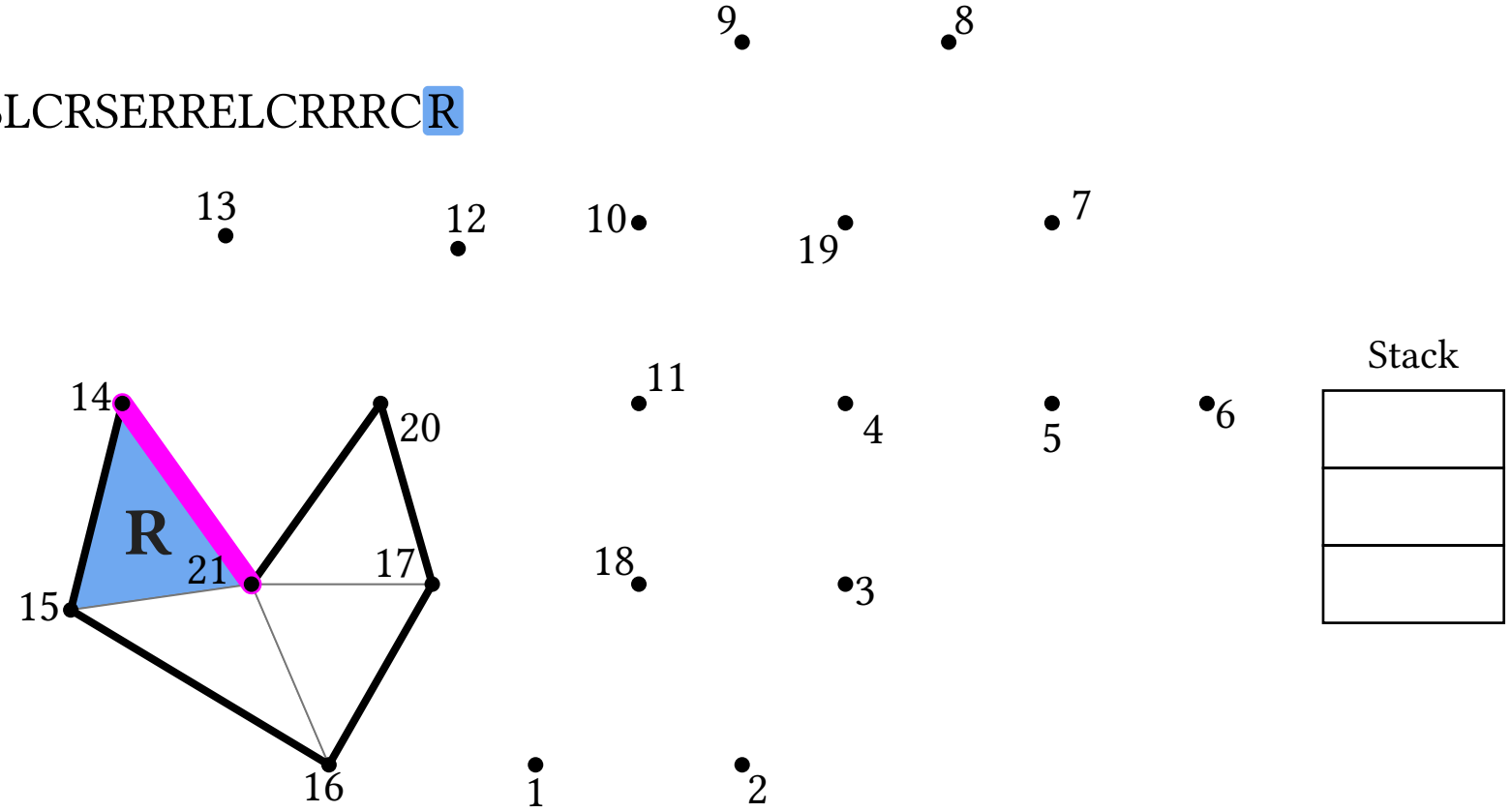


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRR

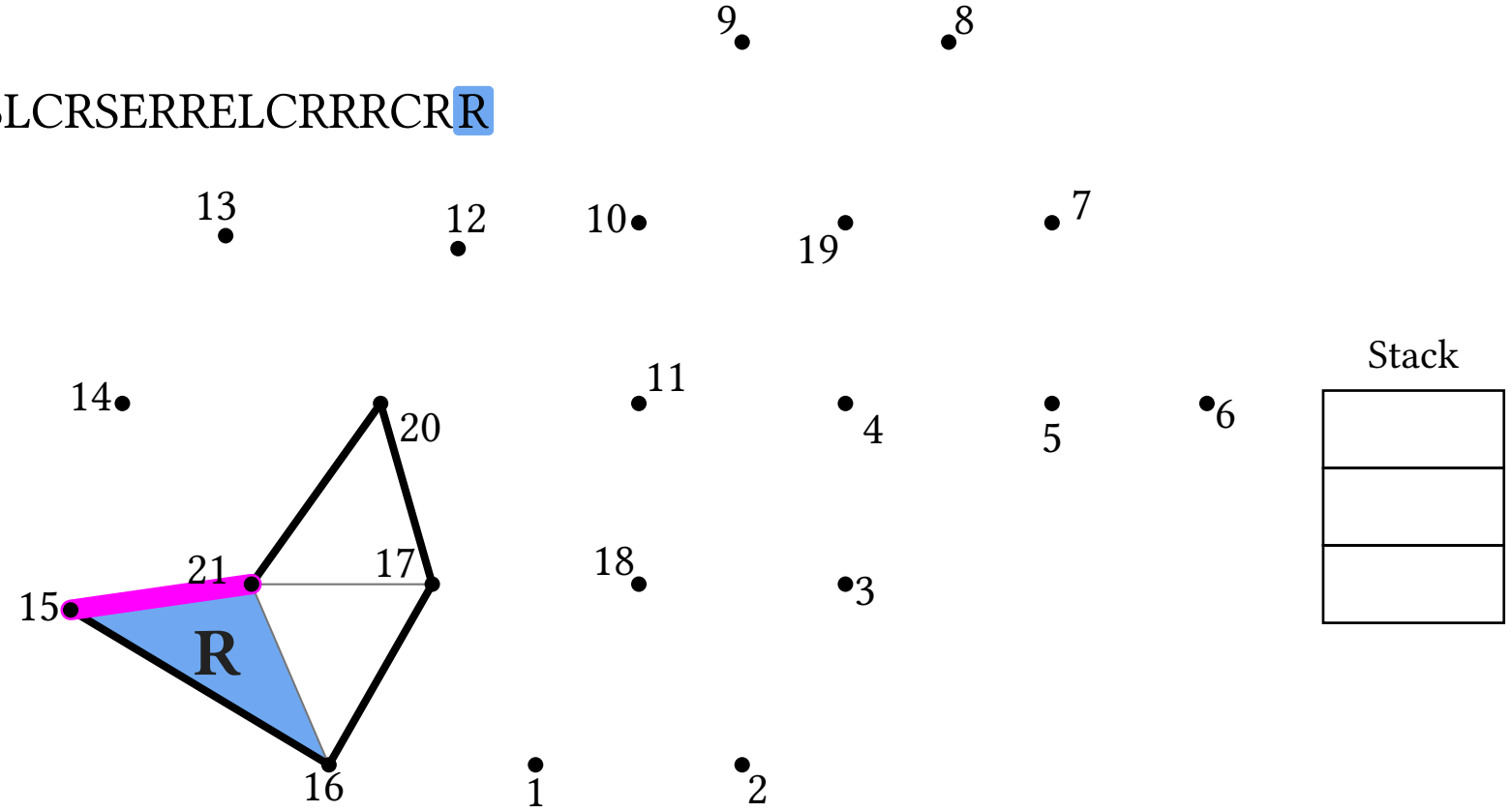


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRR

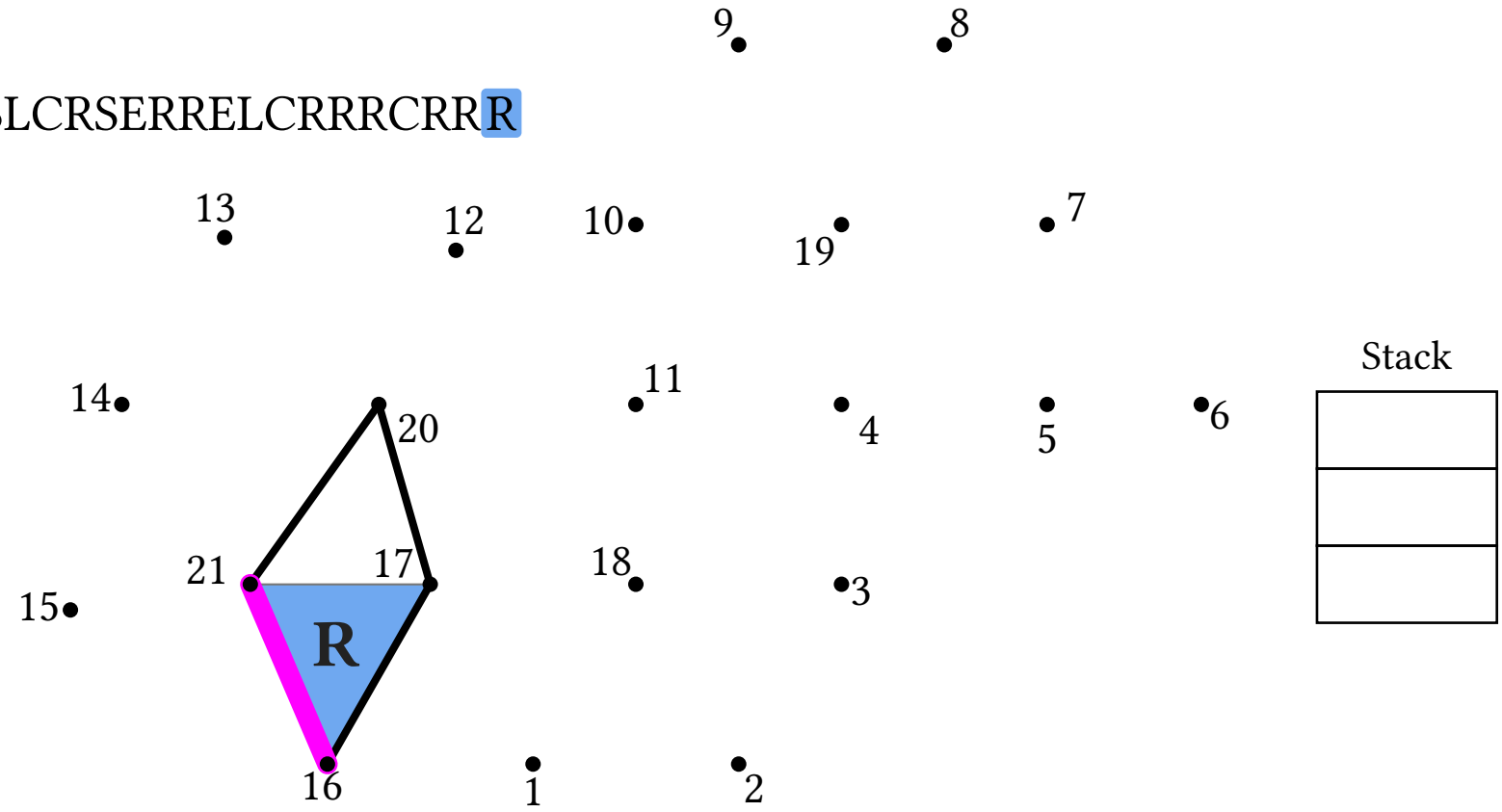


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

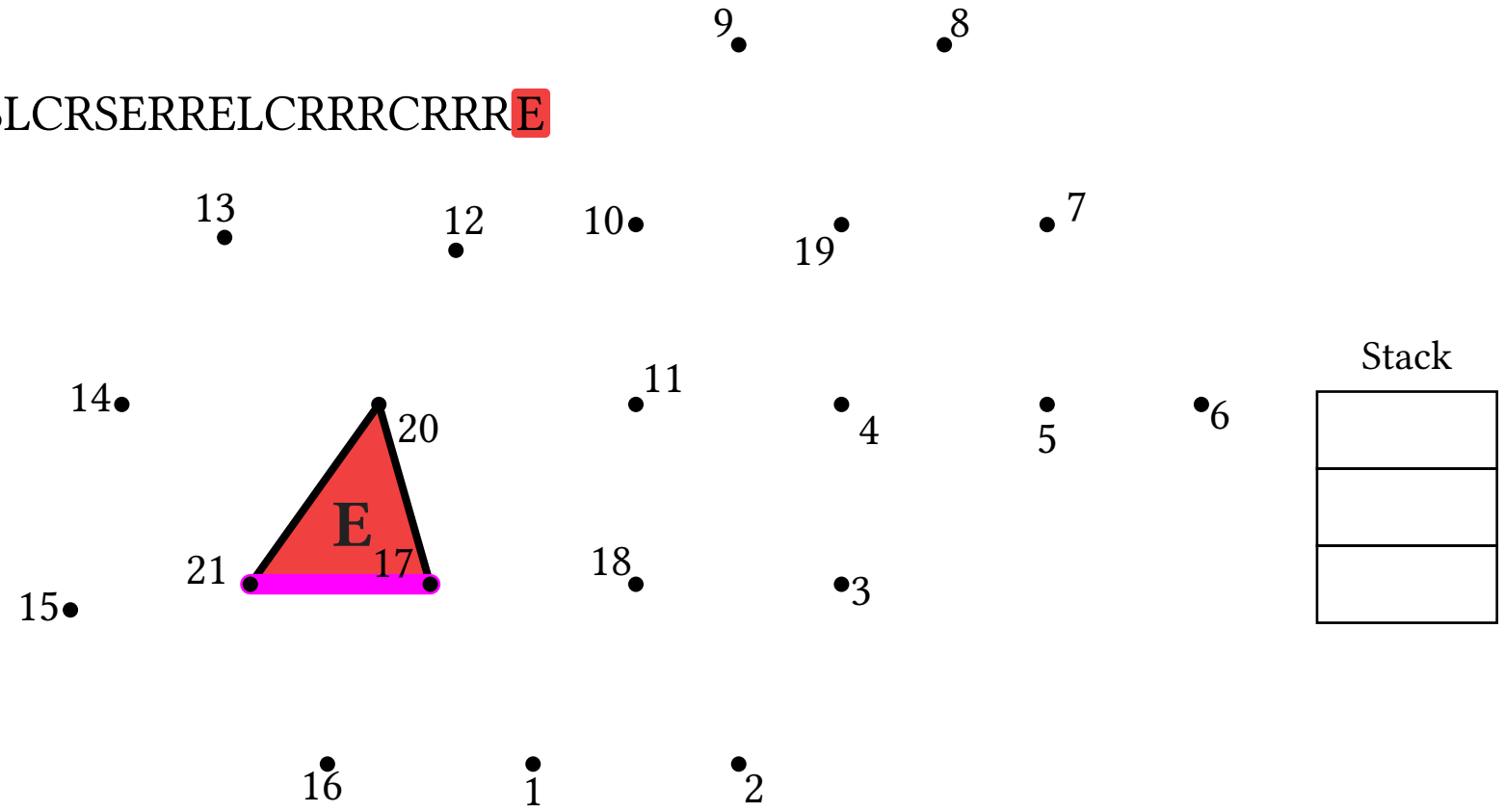


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

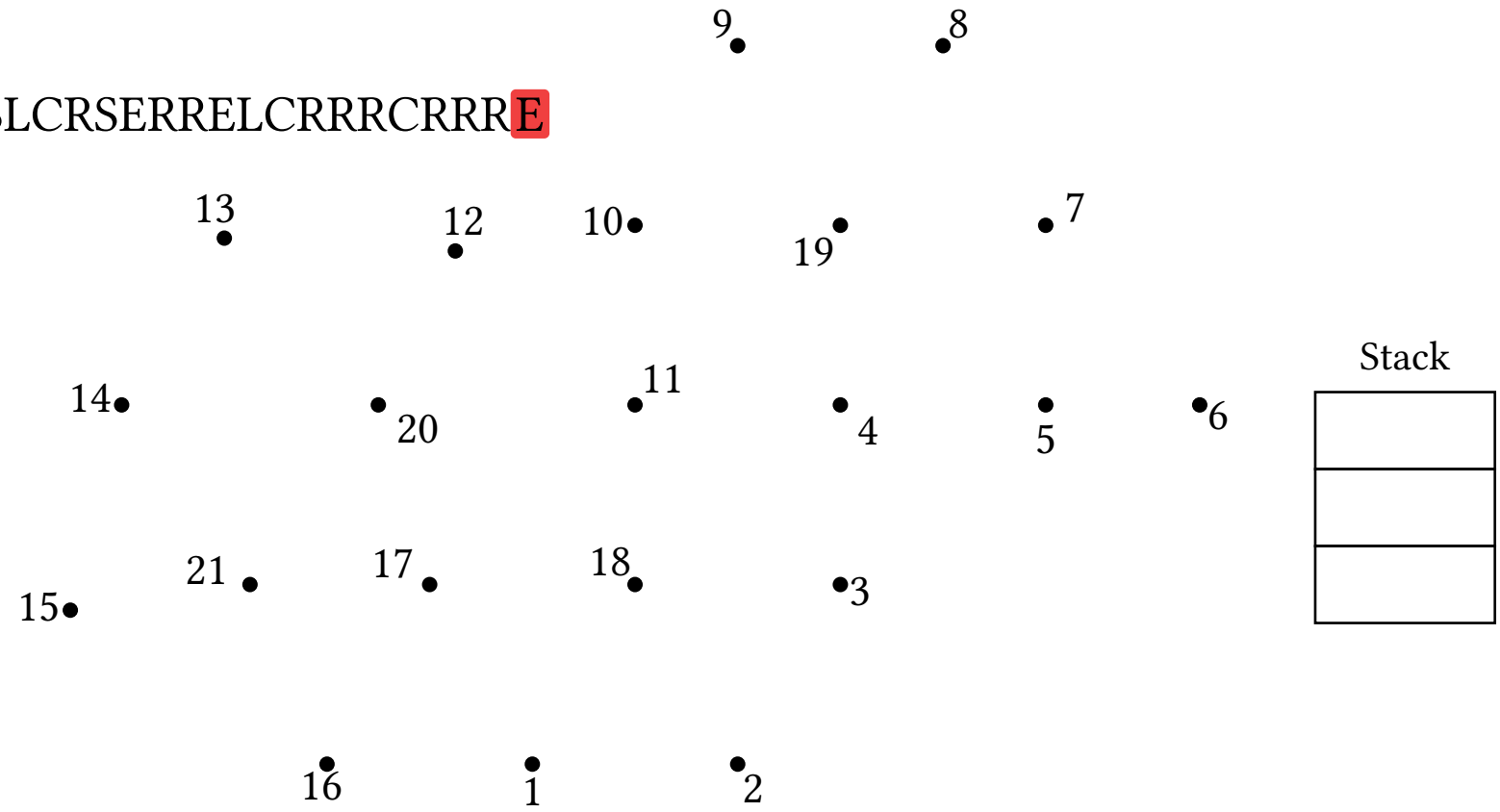


Figure 5: Jednoduchá komprese.

3.3 Vyšší genus?

History =

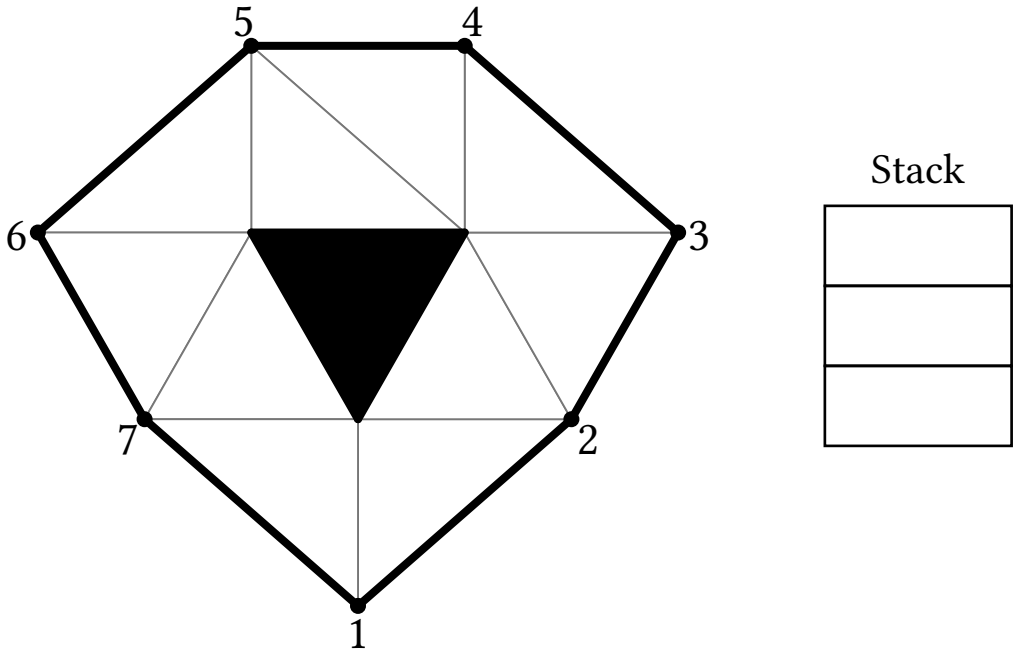


Figure 6: Komprese díry.

3.3 Vyšší genus?

History = S

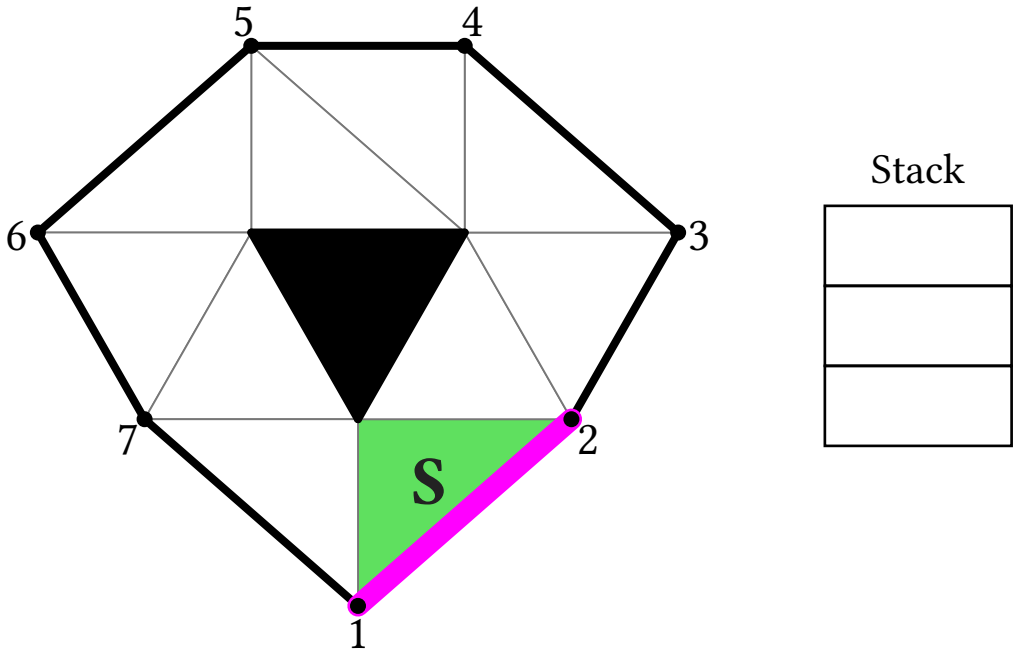
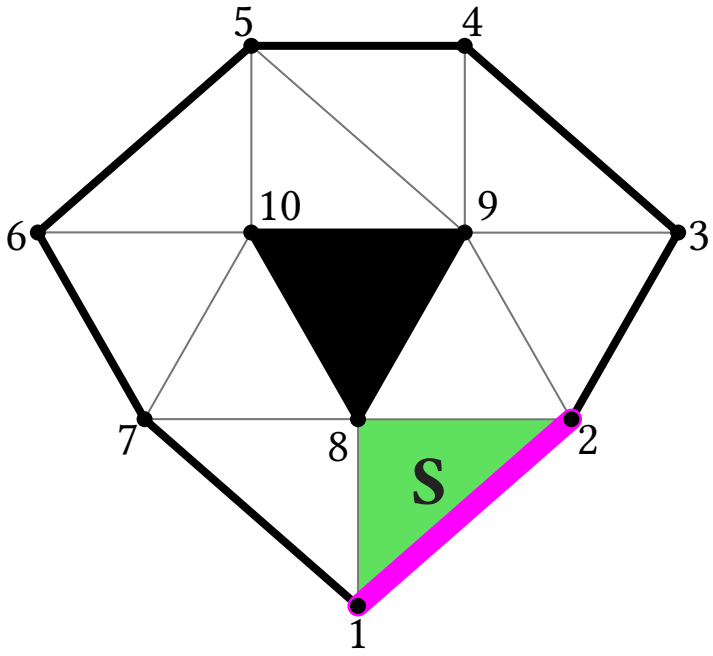


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = S



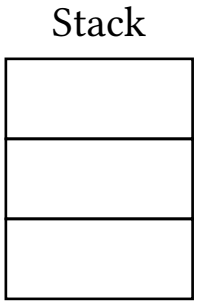


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SL

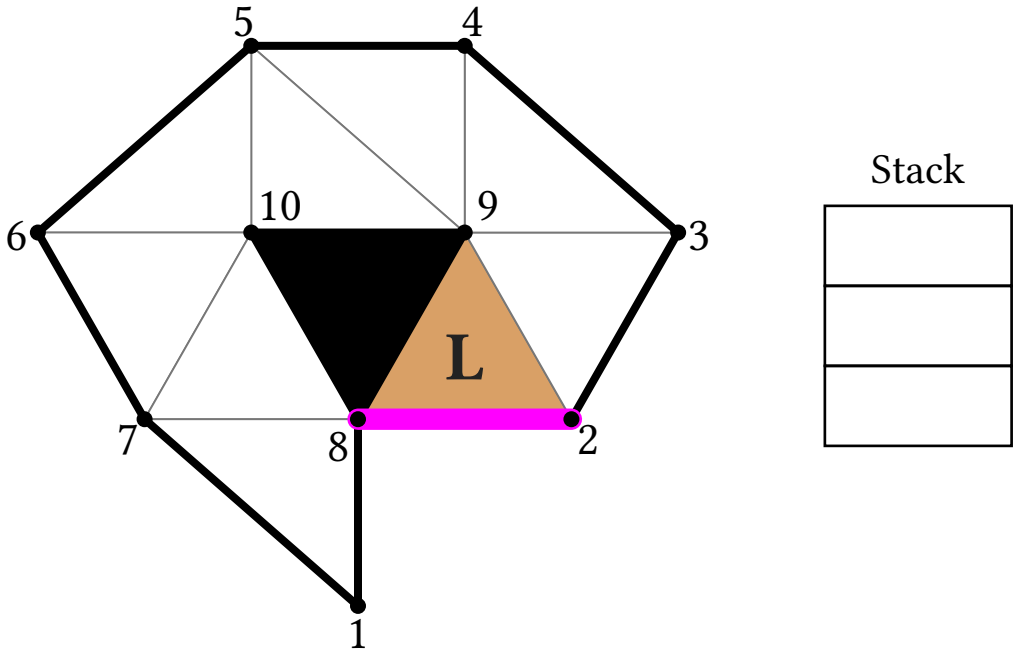


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLR

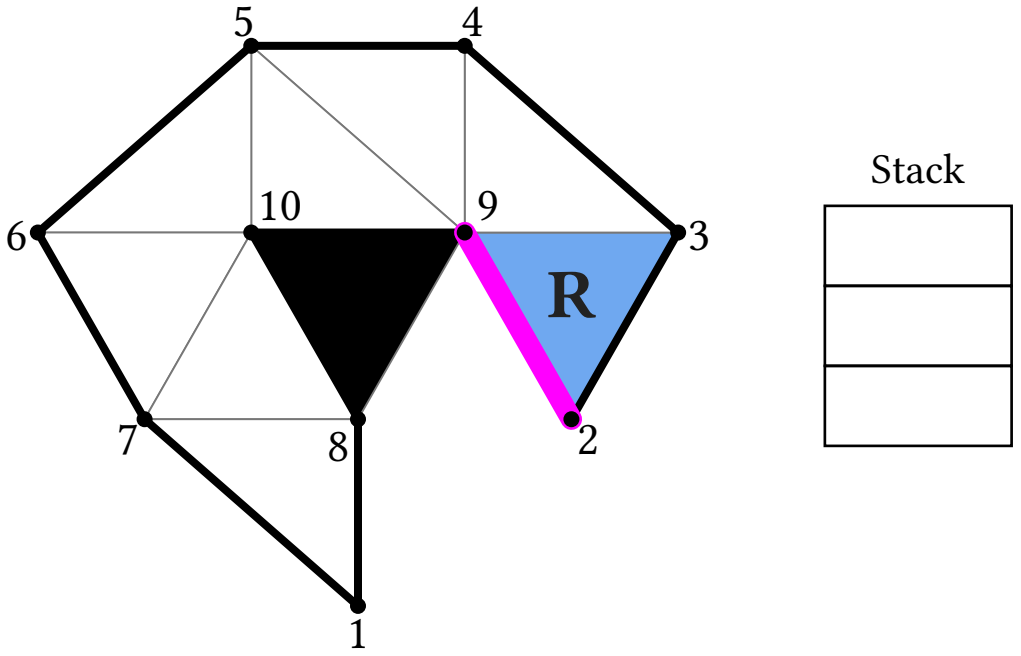


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRR

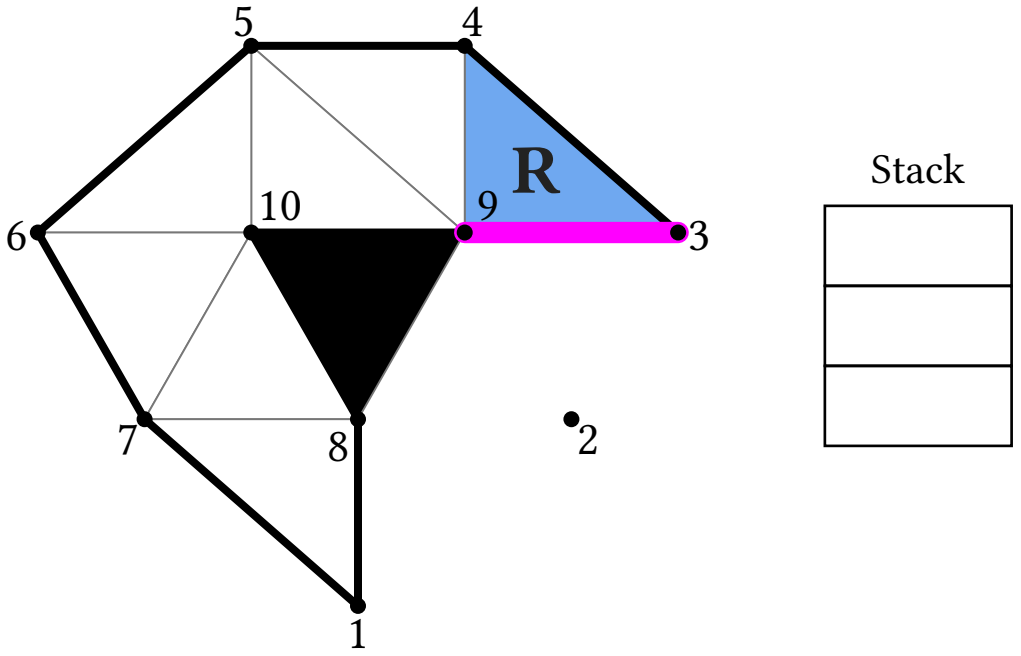


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRR

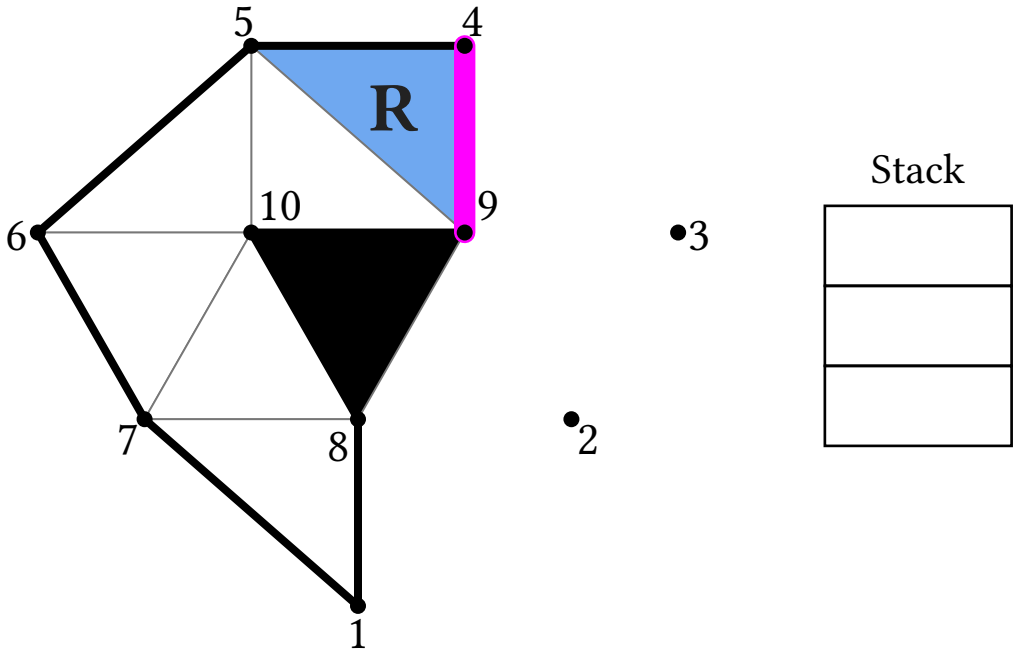


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRL

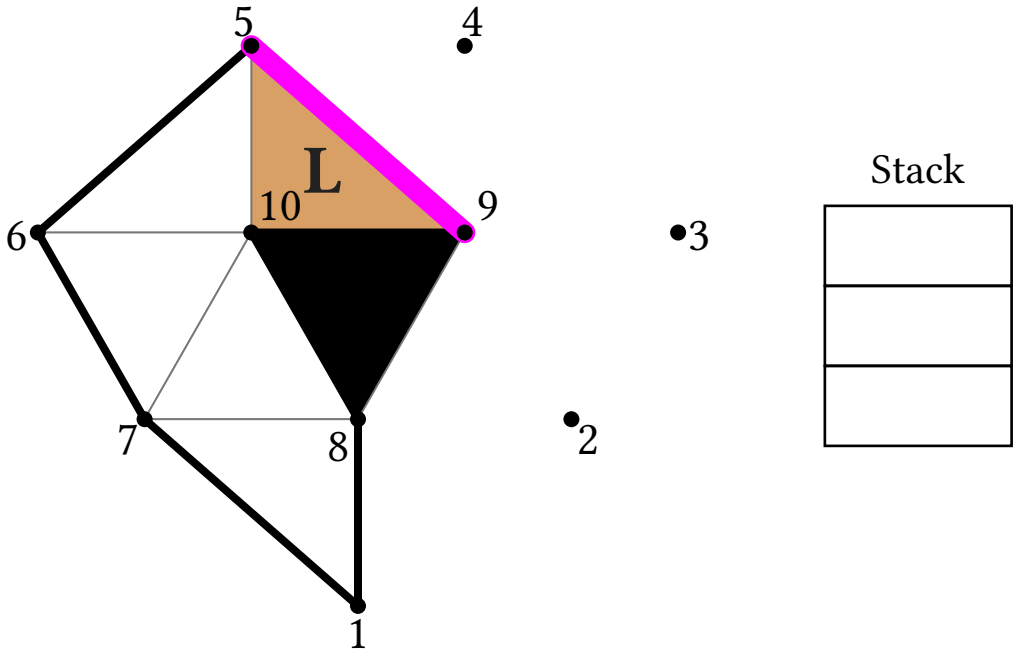


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLR

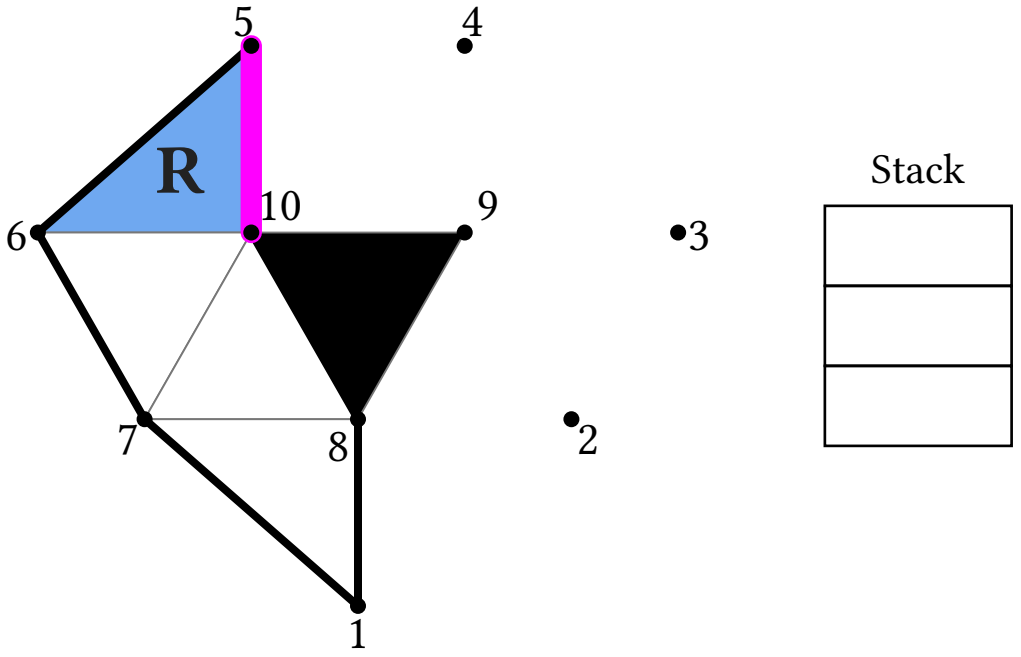


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRR

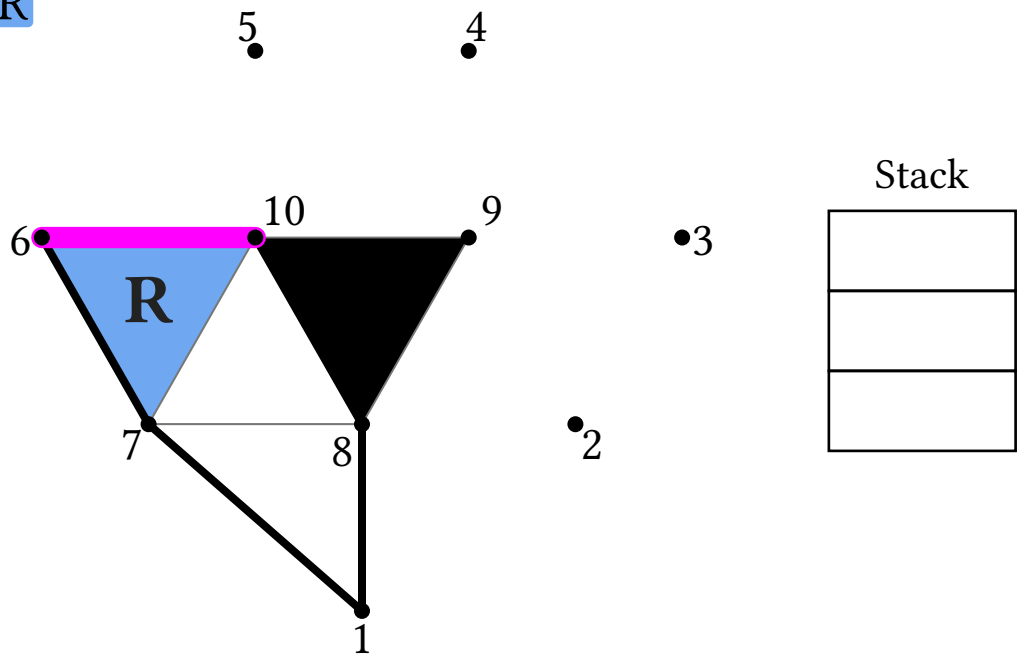


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRL

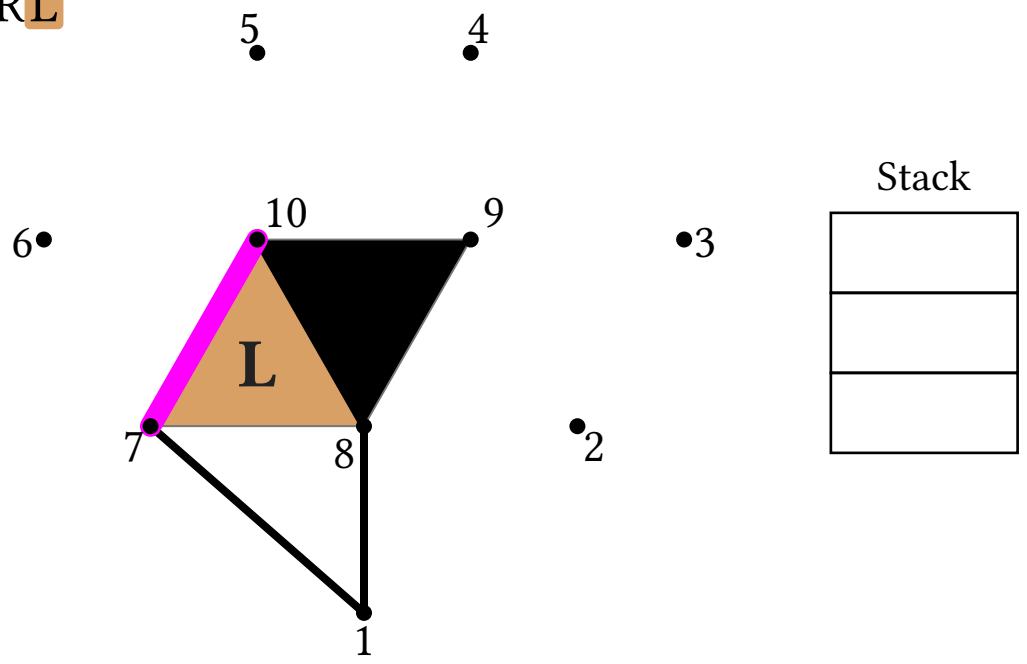


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRLE

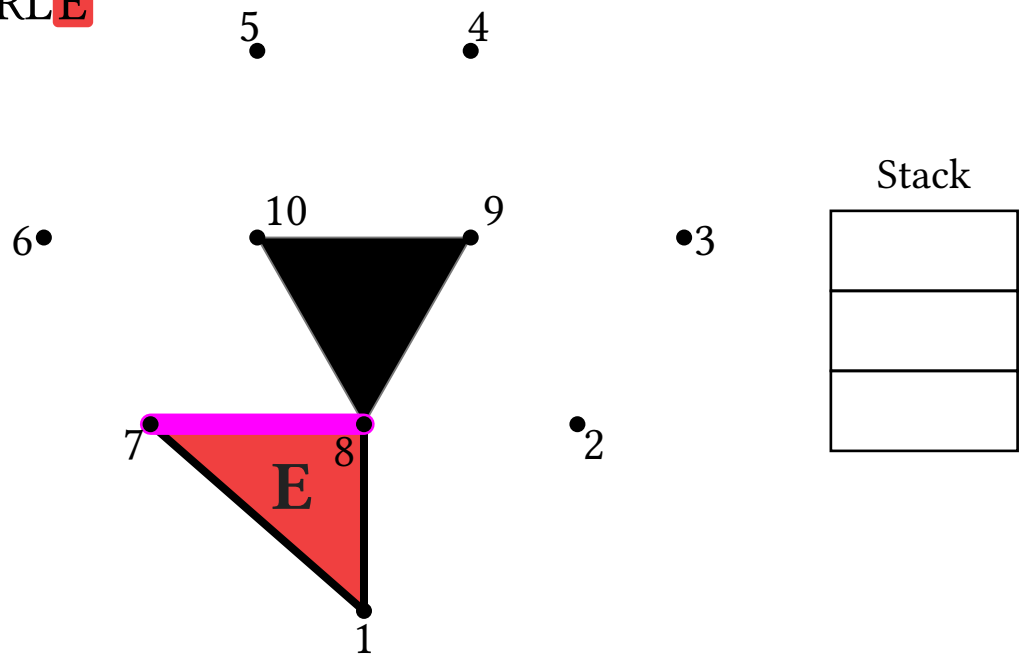


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRLE

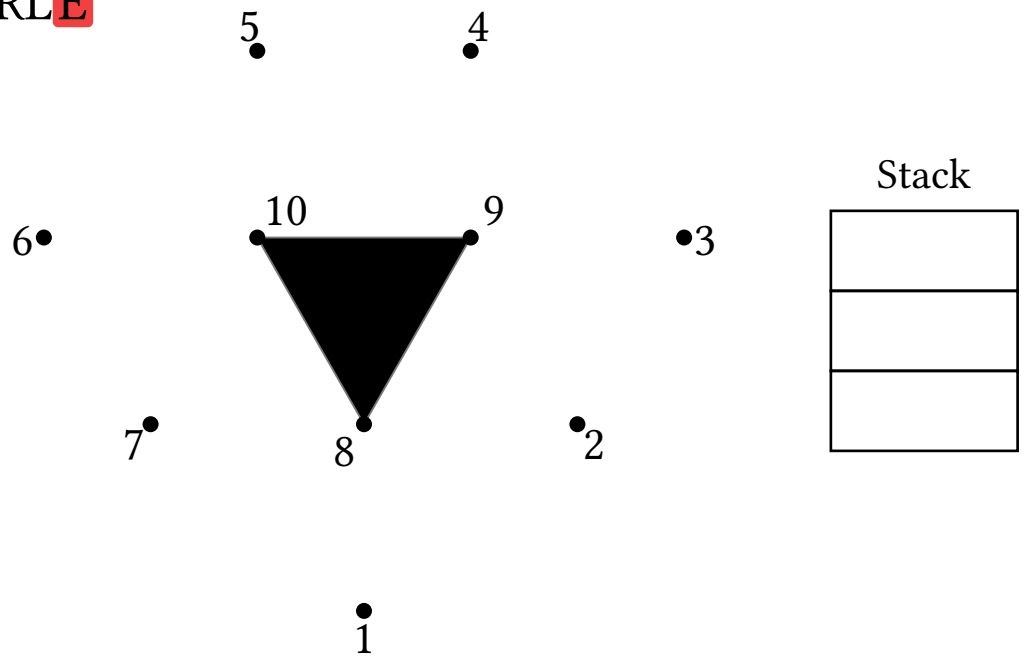


Figure 8: Komprese díry.

4. Dekomprese

4.1 Možnosti

4.2 Pravidla

History = CCRRRSLCRSERRELCRRRCRRRE

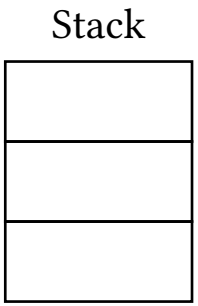


Figure 9: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE

Stack

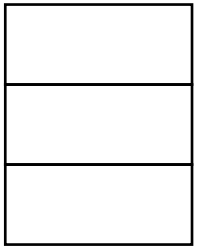


Figure 9: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE



Figure 9: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRR



Figure 9: Dekomprese.

5. Zdroje

5.1 Seznam figures

Figure 1: Dobový kontext vzniku algoritmu Edgebreaker. 2
Figure 2: Jarek Rossignac, zdroj: http://kelvinkuo.com/wp-content/uploads/2010/03/KKK_6258.jpg 3
Figure 3: Trojúhelníková mesh. 5
Figure 4: Základní CLERS operace. 7
Figure 5: Jednoduchá komprese. 10
Figure 6: Komprese díry. 11
Figure 7: Dekomprese. 15

5.2 Obrázky (dobový kontext)

1	**Nagano** (2/1998)	Wikipedia, *fair use*
2	**Duda** (4/1998)	Gov.pl, CC BY 3.0 PL, Wikimedia Commons
3	**Windows 98** (5/1998)	SoundCloud (artwork)
4	**MrBeast** (5/1998)	YouTube (avatar kanálu)
5	**Google** (9/1998)	logodix.com
6	**Half-Life** (11/1998)	Valve, valvearchive.com Wikipedia *fair use*
7	**NATO** (3/1999)	nato.int
8	**Futurama** (3/1999)	ČSFD (plakát)
9	**SpongeBob** (5/1999)	craiyon.com
10	**Napster** (6/1999)	Jusore, GPLv3, Wikimedia Commons
11	**Putin** (12/1999)	Kremlin.ru, CC BY 4.0, Wikimedia Commons

5.3 Reference

Originální paper [1] Corner table [2] Významné vylepšení [3] Wrap & Zip [4] Inspirace animací [5]

[1] J. Rossignac, “Edgebreaker: Connectivity Compression for Triangle Meshes,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 5, no. 1, pp. 47–61, Jan. 1999, doi: 10.1109/2945.764870.

[2] J. Rossignac, A. Safonova, and A. Szymczak, “Edgebreaker on a Corner Table: A Simple Technique for Representing and Compressing Triangulated Surfaces,” in *Hierarchical and Geometrical Methods in Scientific Visualization*, G. Farin, B. Hamann, and H. Hagen, Eds., Berlin, Heidelberg: Springer, 2003, pp. 41–50. doi: 10.1007/978-3-662-07443-5_3.

[3] T. Lewiner, H. Lopes, J. Rossignac, and A. W. Vieira, “Efficient Edgebreaker for Surfaces of Arbitrary Topology,” in *Proceedings of SIBGRAPI 2004 (XVII Brazilian Symposium on Computer Graphics and Image Processing)*, Curitiba, Brazil, 2004, pp. 218–225. doi: 10.1109/SIBGRA.2004.1352965.

[4] J. Rossignac and A. Szymczak, “Wrap&Zip: Linear Decoding of Planar Triangle Graphs,” technical report GIT-GVU-99-08, Feb. 1999.

[5] “Edgebreaker: Connectivity Compression for Triangle Meshes.” Accessed: Nov. 03, 2026. [Online]. Available: https://graphics.stanford.edu/courses/cs468-08-winter/slides/sid2.pdf